

Chapter 1

How Green and Does it Clean: Methodologies for Assessing Cleaning Products for Safety and Performance

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1 BACKGROUND

Every article, chapter, or book on surface cleaning starts with a definition of what clean means, and there is no exception here. An easy way to define it would be to say clean means free from dirt, stain, or impurities, or, even more simply, unsoiled. Cleaning is the process of getting rid of these impurities. These soils can be defined as any extraneous or unwanted material deposited and/or attached to a surface. These are dictionary definitions. In most cases, people ask the basic question of how to determine if something is clean. However, the question should not be “How clean is clean?” but it should be “How clean is clean enough?”

Industrial surface cleaning is performed for a number of reasons. It may be required to prepare the surfaces of parts prior to other fabrication processes such as welding or painting. It may be performed for aesthetic reasons as an aid in marketing and sales. Or it may be necessary to ensure that the finished product will perform without failure caused by contamination. Because of the serious mechanical accidents that can occur as a result of poor surface cleaning and/or inspection, almost all cleaning regimens are developed on an application-specific basis [1].

Much data on cleaning have been gathered by personnel within individual companies, defense contractors, and the military service branches. Some government agencies have been instrumental in compiling these case studies into

databases. Yet the information derived from these sources does not usually reveal the selection process for which chemical cleaners and cleaning equipment were tested and, equally importantly, how they were tested.

2 ENVIRONMENTAL CONCERNS

As important as “How clean is clean enough?” to the cleaning industry should be the question “How safe is cleaning?” Many cleaning products contain harmful chemicals that can have serious adverse effects on cleaning personnel, building occupants, and the environment. These products can cause harm to the environment during their use if they are poured down drains, circulated through building ventilation systems, or disposed of outdoors. Environmental damage can also occur during the development, manufacture, and transport of these products. Possible environmental consequences of cleaning products use include:

- Air pollution
- Bioaccumulation of toxic substances in plants and animals
- Endocrine disruption in wildlife, which reduces the ability to reproduce
- Ozone depletion
- Water pollution.

Industrial chemicals, alone or in combination, have been incorporated into millions of products used every day and are the platform upon which key elements of the global economy have been built. However, many may have inherent characteristics that can harm ecological and human systems as they are used throughout supply chains. Chemicals can be toxic to ecosystems, harm human health, or build up in the environment and in our bodies for decades without breaking down. To identify and control these hazards, to promote safer, more sustainable solutions, it is necessary to manage the ways in which chemicals are used and disposed of [2].

3 GREEN CLEANING

A green cleaning program goes beyond chemical and equipment choices. It includes policies, procedures, training, and shared responsibility efforts that minimize the impact of cleaning materials on the health of building occupants and protect the environment as a whole.

Green cleaning refers to using cleaning methods and products with environmentally friendly ingredients designed to preserve human health and environmental quality. Green cleaning builds on 12 principles of Green Chemistry which are listed in [Table 1.1](#).

Green Chemistry incorporates the disciplines of chemistry and chemical engineering to design chemical products and processes that reduce or eliminate the use or generation of hazardous substances while producing high-quality products through safe and efficient manufacturing processes. Green Chemistry

4 Developments in Surface Contamination and Cleaning

TABLE 1.1 Twelve Principles of Green Chemistry are Outlined in the Book *Green Chemistry: Theory and Practice* [3]

Principle	Description
1	Prevent waste: Design chemical syntheses to prevent waste, leaving no waste to treat or clean up
2	Design safer chemicals and products: Design chemical products to be fully effective, yet have little or no toxicity
3	Design less hazardous chemical syntheses: Design syntheses to use and generate substances with little or no toxicity to humans and the environment
4	Use renewable feedstocks: Use raw materials and feedstocks that are renewable rather than depleting. Renewable feedstocks are often made from agricultural products or are the wastes of other processes; depleting feedstocks are made from fossil fuels (petroleum, natural gas, or coal) or are mined
5	Use catalysts, not stoichiometric reagents: Minimize waste by using catalytic reactions. Catalysts are used in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and work only once
6	Avoid chemical derivatives: Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste
7	Maximize atom economy: Design syntheses so that the final product contains the maximum proportion of the starting materials. There should be few, if any, wasted atoms.
8	Use safer solvents and reaction conditions: Avoid using solvents, separation agents, or other auxiliary chemicals. If these chemicals are necessary, use innocuous chemicals
9	Increase energy efficiency: Run chemical reactions at ambient temperature and pressure whenever possible
10	Design chemicals and products to degrade after use: Design chemical products to break down to innocuous substances after use so that they do not accumulate in the environment.
11	Analyze in real time to prevent pollution: Include in-process, real-time monitoring and control during syntheses to minimize or eliminate the formation of by-products
12	Minimize the potential for accidents: Design chemicals and their forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment

seeks to transform industrial activity to create a zero-waste society. Rather than manage the risks, Green Chemistry seeks to reduce or eliminate them.

The principles that are typically applied to the cleaning industry include:

- Design safer chemicals and products (2)
- Use renewable feedstock (4)
- Increase energy efficiency (9)
- Design chemicals and products to degrade after use (10)
- Minimize the potential for accidents (12).

These may not be an exact match to the industry sector of interest, but the underlying principles can be applied.

Principle 2 capitalizes on the design for a safer environment movement. The United States Environmental Protection Agency (EPA) Design for the Environment (DfE) program was established to help consumers, businesses, and institutional buyers to identify cleaning and other products that perform well, are cost-effective, and are safer for the environment. The DfE program works in partnership with industry, environmental groups, and academia to reduce risk to people and the environment by finding ways to prevent pollution. For more than 15 years, through partnership projects, DfE has evaluated human health and environmental concerns associated with traditional and alternative chemicals and processes in a range of industries. These analyses have empowered hundreds of businesses to select safer chemicals and technologies. DfE focuses on industries that combine the potential for chemical risk reduction with a strong motivation to make lasting, positive changes [4].

New cleaning products are continuously being produced to be more effective, yet have less or lower toxicity than previous generations of products. Just because you tried a green cleaner previously and were unhappy with the results does not mean there will not be something else out there soon that will work.

Principle 4 works toward achieving a sustainable system that will lower the impact of the cleaning industry on the environment. The key aspect of this principle is to find a way to create cleaning products from waste stream agricultural products so as not to disrupt food sources being diverted to manufacturing of products. Some of these products include solvents based on renewable feedstock: biopolyesters such as polyhydroxyalkanoates, methyl soyates, ethyl lactates, and citrus terpenes/oils. The bio-based alternatives perform fairly well on many of the metalworking fluids. These products deliver cleaning efficiencies comparable to aqueous-based formulations, petroleum distillates, and trichloroethylene (TCE). The D-limonene-based products tend to be better suited for the heavier metalworking fluids like the greases and the rust preventative coating. For many of the soy-based products, film residue decreased the initial efficiencies. By altering the rinsing (spray) and/or drying (high velocity air blow off or wiping) methodologies, this residue could be reduced or eliminated. For the nonmetal working soils, the D-limonene products had a wider range of success than the soy and lactate products [5].

Principle 9 relates to the cleaning industry through products that can be used at low temperatures and require less mechanical energy (vibration, ultrasonic, etc.) to achieve desirable results.

Principles 10 and 12 showcase how many of the cleaning products on the market have been designed to degrade after use and to minimize hazards during normal usage.

4 REGULATORY ASPECTS

Solvents have been used for many years in all fields of cleaning. Many of these solvents, even though they work well, pose health risks to workers and are regulated by state and federal agencies. Efforts have been made to reduce worker exposure. More often than not, these changes arise as a result of a company trying to adhere to the various legal structures that exist, especially liability issues, and account for how most societies approach environmental decision making for cleaning applications. Therefore, as one tries to follow the list of myriad regulations and restrictions, identifying a substitute for solvents in cleaning applications is not an easy task. There are literally thousands of formulations to choose from. The information and literature for these products changes from vendor to vendor as well as from product to product, making the search for an applicable substitute nearly impossible. Even after selecting a potential product, there is no guarantee that it will work [6].

In the cleaning industry, regulations are always changing. Attempting to discuss the current regulatory requirements would only be a snapshot in time, quickly becoming a time capsule of what was once the regulatory environment. There are many good interactive tools available to help those in search of the current rules and restrictions.

One such tool is the State Chemicals Policy Database. This tool is a searchable database of passed and pending state-level chemical policy legislation originally developed by the Lowell Center for Sustainable Production. Users can search the database by state, region, status (i.e., enacted, proposed, or failed), policy category (e.g., pollution prevention, single chemical restriction, and others), chemical, and product type (e.g., children's products, cleaning products, and others) [7].

Legislation that is included in the database falls into 13 policy categories: (1) pollution prevention and toxics use reduction (TUR); (2) single chemical restrictions; (3) multiple chemical policies; (4) regulation of product categories; (5) biomonitoring and environmental health tracking and surveillance systems; (6) data collection; (7) right-to-know; (8) prioritization; (9) alternatives assessment; (10) green chemistry and DfE; (11) product stewardship; (12) environmentally preferable purchasing; and (13) precautionary principle and are listed in [Table 1.2](#).

The database can be searched by using the seven pull-down menus (i.e. state, region, status, chemical, policy category, product types, year) to make selections.

TABLE 1.2 Policy Examples Included in State Chemicals Policy Database Tool^a [8]

Policy Category	Definition	Policy Landscape at State/Local Levels	Examples
Pollution prevention/toxics use reduction	Multipollutant, multimedia strategies that shift the focus from end-of-pipe regulation to reduction of pollution at the source and/or encourage changes in production processes, product, or raw materials to reduce, avoid, or eliminate the use of toxic or hazardous substances or the generation of hazardous by-products	Following federal pollution prevention legislation in 1990, many states enacted similar pollution prevention laws. Although the majority of states have set pollution prevention goals, there is a wide variety of programs and policies set up to achieve these goals at the state level	Enacted-Massachusetts Toxic Use Reduction Act (TURA); Oregon Toxics Use Reduction and Hazardous Waste Reduction Act; New Jersey Pollution Prevention Act Proposed—New York (A348, S2256); California (AB558)
Single chemical restrictions	Policies that ban or severely restrict specific chemicals or uses of chemicals	Prominent at the state and local levels. A number of chemicals have been banned or restricted in some states and localities. In addition, there is proposed legislation in a number of states and localities to ban or restrict single chemicals. The chemicals targeted by these policies include: PBDEs, lead, mercury, chromated copper arsenate, chlorinated solvents, dioxin, formaldehyde, perchloroethylene, phthalates, bisphenol A, and diacetyl	Enacted-Washington polybrominated diphenyl ether (PBDE) Ban (Wash. Rev. Code Ann. §70.76.005); Maine PBDE Ban (Me. Rev. Stat. Ann. tit. 38, § 1609); Rhode Island Mercury Reduction and Education Act

Continued

TABLE 1.2 Policy Examples Included in State Chemicals Policy Database Tool [8]—Cont'd

Policy Category	Definition	Policy Landscape at State/Local Levels	Examples
Right-to-know	Policies that require or encourage the provision of information or disclosures about exposures and health risks associated with chemicals to the general public	Some right-to-know policies at the state and local level build on existing federal legislation (Emergency Planning and Community Right-to-Know Act) and require users of hazardous substances to publicly report these uses	Enacted-California Safe Drinking Water and Toxic Enforcement Act (Prop. 65); California Toxics Information Clearinghouse (SB509); City of Eugene, Oregon Toxics Right-to-Know Charter Amendment; New Jersey Worker and Community Right-to-Know

^aSpecific areas that are excluded from the database include legislation related to: pesticides, pharmaceuticals, media-specific initiatives (air, water, soil, sludge, etc.), mercury in vaccines, and cleanup/remediation/site-specific initiatives.

When making selections in multiple pull-down menus, the results will include only entries that contain all of the selections highlighted. The database can also be searched by making multiple selections from one pull-down menu. When making multiple selections in one pull-down menu, the results will include entries that contain any of the selections highlighted.

Additionally, the entire database can be searched by entering a bill number, word, or phrase into the box located below the pull-down menus. The database layout is shown in [Fig. 1.1](#). This will search the full database entries of each policy for the entered word or phrase, although it will not return results where the entered word or phrase is found solely in the full-text document of the policy [10].

5 PRODUCT SELECTION

Normally, the first thing someone thinks of for an effective cleaning product is does it work. If you are lucky enough to find something that works great, you probably then walked head first into the regulatory wall, realizing what you just picked is on a “Do Not Use” list for any one of the governing environmental enforcement bodies (air, water, and solid waste). How do you get to a product that works and is safer? But if you come at from the environmental, health, and

STATE	REGION	STATUS	CHEMICAL
All	Midwest	All	All
Alabama	Northeast	Proposed	2-ethyl-1-hexanol
Alaska	Pacific	Enacted	4-phenylecyclohexene
Arizona	Southeast	Failed	Alkylphenol
Arkansas	Southwest		Arsenic
California	West		benzyl butyl phthalate (BBP)

POLICY CATEGORY	PRODUCT TYPES	YEAR
All	All	All
Alternatives Assessment	Brake Friction Materials	2013
Biomonitoring	Children's Products	2012
Data Collection	Cleaning Products	2011
Design for the Environment	Clothing and Footwear	2010
Environmental Health Tracking and Surveillance Systems	Cosmetics	2009

SEARCH ENTIRE DATABASE

Enter words or phrases, separated by commas.

SEARCH

FIGURE 1.1 Policy database search options [9].

safety (EH&S) direction, you might find a product that is safer to use but when you go to use it, the product does not work well for your needs. The best approach is to use a screening system to eliminate really hazardous products, and then test for performance of the potentially safer products. Once you find a few that work well, you then need to complete a more comprehensive safety evaluation of the products.

So, before you can start testing products for performance, you need to select products for evaluation. The product selection process needs to first address laws and restrictions if you want to stay ahead of the regulatory matching game. From there, the products must then be reviewed from a health and safety comparison and finally, they need to be evaluated for efficacy as it pertains to your needs.

6 GREEN ASSESSMENT—WHAT MAKES A PRODUCT GREEN?

Performance is not the only selection criterion that needs to be addressed when looking for a new cleaning product. Traditional solvents have known EH&S effects, which has led to the development of alternative chemical formulations. The toxicity, flammability, and smog-producing effects of petroleum-based solvents were recognized from an early date to be an area of concern. Chlorofluorocarbons (CFCs) were designed for the replacement of the petroleum solvents

but, despite being some of the most useful classes of compounds ever synthesized, the chemical stability of CFCs caused long atmospheric residence times. This allowed CFCs to slowly mix in the upper atmosphere where they dissociate, freeing chlorine atoms, and catalyzing the destruction of the Earth's protective ozone layer.

The most important consideration when choosing an alternative is not to shift the risk involved in the process. For example, when attempting to replace a flammable solvent, it would not be advantageous to select an ozone-depleting chemical. In this case, the risk goes from the worker (burn, explosion, etc.) to the environment (ozone destruction). Ideally, a selection of an alternative cleaning product should be safer for the worker or the environment, or, if possible, a product that would result in a reduced risk for both the worker and the environment.

6.1 Pollution Prevention Options Assessment Tool (P2OASys)

The Toxics Use Reduction Institute (TURI) at the University of Massachusetts Lowell provides resources and tools to help make the Commonwealth a safer and more sustainable place to live and work. Established by the Massachusetts Toxics Use Reduction Act (TURA) of 1989, TURI collaborates with businesses, community organizations, and government agencies to reduce the use of toxic chemicals, protect public health and the environment, and increase competitiveness of Massachusetts businesses.

In the mid-1990s, the Institute developed a systematic tool to help companies determine whether the toxics use reduction (TUR) options they were considering may have unforeseen negative environmental, worker, or public health impacts. P2OASys allows companies to assess the potential environmental, worker, and public health impacts of alternative technologies aimed at reducing toxics use. The goal is more comprehensive and systematic thinking about the potential hazards posed by current and alternative processes identified during the TUR planning process [11].

The tool assists companies in two ways:

- Systematic examination of the potential environmental and worker impacts of TUR options in a comprehensive manner, examining the impacts of process changes, rather than simply those of chemical changes.
- Comparison of TUR options with the company's current process based on quantitative and qualitative factors.

The tool employs 61 elements associated with EH&S. The elements are divided into 11 major categories. These include: acute human effects, chronic human effects, physical hazards, aquatic hazards, persistence/bioaccumulation, atmospheric hazard, disposal hazard, chemical hazard, energy and resource use, product hazard, and exposure potential.

The scoring system uses a 10-point scale with 2 being the best score and 10 being the worst. For criteria not assigned a numerical input, the tool allows for an expert judgment to be made on a low, medium, or high rating system denoted by L, M, or H, respectively.

Formulae embedded in P2OASys provide a numerical hazard score for the company's current process and identified options, which can then be combined with other information sources and professional expertise to make decisions on TUR option implementation. [Table 1.3](#) lists the categories and the evaluation criteria used in the assessment process [12].

The P2OASys tool provides the user with information on safety for both workers and the environment. It is not designed to propose an alternative cleaning product as the user must provide the alternatives compared in this tool. The program can take time to use, as well as considerable effort to populate the matrix with alternative product information. However, once the data are entered, the tool provides for a consistent method to compare current solvents used with possible alternatives. One of the benefits of this tool is the ability to use both quantitative and qualitative data on the chemical toxicity, ecological effects, and physical properties of both current and proposed chemistries [13].

6.2 Green Screen

Clean Production Action designs and delivers strategic solutions for green chemicals, sustainable materials, and environmentally preferable products. Their role is to translate the systems-based vision of clean production into the tools and strategies that non-government organizations (NGOs), governments, and businesses need to advance green chemicals, sustainable materials, and environmentally preferable products [14].

The Clean Product Action's Green Screen for Safer Chemicals defines a path to chemicals that are safer for humans and the environment. It is intended to be a rigorous, hazard-based screening method that is designed to inform decision making by businesses, governments, and individuals concerned with the risks posed by chemicals and to advance the development of green chemistry [15]. The elements of the Green Screen process are similar to those in the P2OASys tool. Of the 19 total elements in the Green Screen, 15 of these appeared in the P2OASys system [13].

Benchmarks are unique to Green Screen and are a major strength of the method. Each benchmark is determined by analyzing specific combinations of hazard classifications. The benchmarks were developed to reflect hazard concerns that have been established by governments nationally and internationally. Each hazard classification is used in applying the benchmark criteria to determine the final benchmark score. It is possible that a benchmark cannot be assigned at all if the data are insufficient [16].

The four benchmarks on the path to safer chemicals are listed here.

TABLE 1.3 P2OASys Criteria and Assessment Levels

Category	Key Criteria	Units ^a	2.00	4.00	6.00	8.00	10.00
Acute human effects	Inhalation median lethal concentration (LC ₅₀)	ppm	10,000	1000	150	15	<15
	Permissible exposure level (PEL)/threshold limit value (TLV)	ppm	200	100	25	5	<5
	PEL/TLV (dusts/particles)	mg/m ³	10	5	1	0.1	<0.1
	Immediately dangerous to life or health (IDLH)	ppm	1000	500	50	10	<10
	Respiratory irritation	L/M/H	L	L/M	M	M/H	H
	Oral median lethal dose (LD ₅₀)	mg/kg	5000	500	50	5	<5
	Dermal irritation	L/M/H	L	L/M	M	M/H	H
	Skin absorption	L/M/H	L	L/M	M	M/H	H
	Dermal LD ₅₀	mg/kg	5000	500	50	5	<5
	Ocular irritation	L/M/H	L	L/M	M	M/H	H
Chronic human effects	Reference dose (RfD)	mg/kg/day	0.1	0.05	0.01	0.001	<0.001
	Carcinogen	International Agency for Research on Cancer (IARC)/EPA	4,E	3,D	2B,C	2A,B	1,A
	Mutagen	L/M/H	L	L/M	M	M/H	H
	Reproductive effects	L/M/H	L	L/M	M	M/H	H
	Neurotoxicity	L/M/H	L	L/M	M	M/H	H
	Developmental effects	L/M/H	L	L/M	M	M/H	H
	Respiratory-sensitivity/disease	L/M/H	L	L/M	M	M/H	H
	Other chronic organ effects	L/M/H	L	L/M	M	M/H	H

Physical hazards	Heat	WBGT, °C	25	27	30	32	>32
	Noise generation	dBA	80	85	85	90	>90
	Vibration	m/S ²	4	6	8	12	>12
	Ergonomic hazard	L/M/H	L	L/M	M	M/H	H
	Psychosocial hazard	L/M/H	L	L/M	M	M/H	H
Aquatic hazards	Water Quality Criteria (HWQC)	mg/L	>10	6-8	4-6	1-4	<1
	Aquatic LC ₅₀	mg/L	1000	50	1	0.1	<0.1
	Fish no observed adverse effect concentration (NOAEC)	mg/L	0.2	0.02	0.002	0.0002	<0.0002
	Plant Half Maximal Effective Concentration (EC ₅₀)	mg/L	100	10	1	0.1	<0.1
	Observed ecological effects	L/M/H	L	L/M	M	M/H	H
Persistence/ bioaccumulation	Persistence	L/M/H	L	L/M	M	M/H	H
	Biochemical oxygen demand (BOD) half-life	days	4	10	100	500	>500
	Hydrolysis half-life	days	4	10	100	500	>500
	Bioconcentration (octanol-water partition coefficient)	log k _{ow}	1	2	4	6	>6
	Bioconcentration factor (BCF)	kg/L	10	100	200	1000	>1000

Continued

TABLE 1.3 P2OASys Criteria and Assessment Levels—Cont'd

Category	Key Criteria	Units	2.00	4.00	6.00	8.00	10.00
Atmospheric hazard	Greenhouse gas	Y/N	N				Y
	Ozone depletion potential (ODP)	ODP units	Not listed				Class I/II
	Acid rain formation	Y/N	N				Y
	National Emission Standards for Hazardous Air Pollutants (NESHAP)	Y/N	N				Y
Disposal hazard	Landfill	L/M/H	L	L/M	M	M/H	H
	Emergency Planning and Community Right-to-Know Act (EPCRA) reportable quantity	lbs	5000	1000	100	10	1
	Incineration	L/M/H	L	L/M	M	M/H	H
	Recycling	L/M/H	L	L/M	M	M/H	H
Chemical hazard	Vapor pressure	mm Hg	0.10	1	10	100	>100
	Solubility in water	mg/L					
	Specific gravity						
	Flammability	0,1,2,3,4	0.0	1	2	3	4
	Flash point	°C	100	75	25	10	<10
	Reactivity	0,1,2,3,4	0.0	1	2	3	4

	pH	pH units	7	6-7, 7-8	5-6, 8-9	3-5, 9-11	1-3, 11-14
	Corrosivity	L/M/H	L	L/M	M	M/H	H
	High-pressure system	L/M/H	L	L/M	M	M/H	H
	High-temperature system	L/M/H	L	L/M	M	M/H	H
	Mixture/reaction potential	L/M/H	L	L/M	M	M/H	H
	Odor threshold	L/M/H	L	L/M	M	M/H	H
	Volatile organic compound (VOC)	L/M/H	L	L/M	M	M/H	H
Energy & resource use	Non renewable resource	L/M/H	L	L/M	M	M/H	H
	Water use	L/M/H	L	L/M	M	M/H	H
	Energy use	L/M/H	L	L/M	M	M/H	H
Product hazard	Upstream effects	L/M/H	L	L/M	M	M/H	H
	Consumer hazard	L/M/H	L	L/M	M	M/H	H
	Disposal hazard	L/M/H	L	L/M	M	M/H	H
Exposure potential	Exposure potential	L/M/H	L	L/M	M	M/H	H

^appm, part per million; mg/m³, milligram per cubic meter; mg/kg, milligram per kilogram; mg/L, milligram per liter; mm Hg, millimeter of mercury; m/s², meter per second squared; Kow, octanol/water partition coefficient; lbs, pounds; dBA, A-weighted decibels; WBGT, Wet Bulb Globe Temperature; IARC, International Agency for Research on Cancer.

- Benchmark 1—Avoid—chemicals of high concern.
- Benchmark 2—Use but search for safer substitutes.
- Benchmark 3—Use but still opportunity for improvement.
- Benchmark 4—Prefer—safer chemicals.

The criteria for each benchmark become progressively more demanding for environmental and human health and safety, with the hazard criteria of Benchmark 4 representing the most preferred chemicals. The hazard level of these criteria will determine in which benchmark the chemical falls into. The more hazardous a chemical, the lower the benchmark the chemical will belong in.

6.2.1 *Human Health Group I*

The endpoints contained in this benchmark reflect priorities that are consistent with national and international governmental regulations and cover hazards that can lead to chronic or life-threatening effects or adverse impacts that are potentially induced at low doses and transferred between generations.

6.2.2 *Human Health Group II and II**

These endpoints reflect hazards that are also important for understanding and classifying chemicals. Typically, Group II hazards may be mitigated. Group II and II* are differentiated from one another in the benchmarking system because Group II endpoints have four hazard levels (i.e., vH, H, M, and L) while Group II* endpoints have three hazard levels (i.e., H, M, and L) and are evaluated based on repeated exposure.

6.2.3 *Environmental Toxicity and Fate*

The environmental toxicity and fate category is focused on the effects of manufactured chemicals and other anthropogenic and natural materials and activities on aquatic organisms. It includes short term (acute), either from a single exposure or from multiple exposures in a short span of time, and prolonged repeated exposures (chronic), often at lower levels, to a substance over a longer time period. In addition, persistence (how long it remains in the environment) and bioaccumulation (how pollutants enter a food chain and increase in concentration) potential. Additional, ecotoxicity endpoints are included when available and relevant.

6.2.4 *Physical Hazards*

Physical hazards include flammability and reactivity and are based on globally harmonized system (GHS) criteria. [Table 1.4](#) lists more specifics on the various groupings and endpoints.

TABLE 1.4 Groupings of Green Screen Hazard

Grouping	Hazard Endpoints
Human health group I	Carcinogenicity (C) Mutagenicity and Genotoxicity (M) Reproductive toxicity (R) Developmental toxicity, including Neurodevelopmental Toxicity (D) Endocrine activity (E)
Human health group II	Acute toxicity (AT) Systemic toxicity and organ effects (ST-single) Neurotoxicity (N-single) Skin irritation (IrS) Eye Irritation (IrE)
Human health group II*	Systemic toxicity and organ effects, repeated exposure sub-endpoint (ST-repeat) Neurotoxicity—repeated exposure sub-endpoint (N-repeated) Skin sensitization (SnS) Respiratory sensitization (SnR)
Environmental toxicity and fate	Acute aquatic toxicity (AA) Chronic aquatic toxicity (CA) Persistence (P) Bioaccumulation (B)
Physical hazards	Reactivity (Rx) Flammability (F)

6.2.5 Green Screen Process Steps

The Green Screen assessment is conducted using the following steps:

Step 1—Determine chemicals to assess

Step 2—Research and collect data

Assessing chemicals is accomplished by examining comprehensive toxicological data, checking Green Screen specified lists, and using estimated data from suitable analogs or modeled data where measured data are lacking for the parent chemical.

Step 3—Classify hazard level for each hazard endpoint (e.g., vH, H, M, L, vL)

Use the hazard criteria resource to classify the hazard level as very High (vH), High (H), Moderate (M), Low (L) or in some cases very Low (vL) for each hazard endpoint. A color scheme is recommended for coloring the box (different gray shades in print version) containing the hazard classification for each hazard endpoint as shown in [Fig. 1.2](#).

Hazard score	
Very High	vH
High	H
Moderate	M
Low	L
Very Low	vL
Data gap	DG

FIGURE 1.2 Color assessment for Green Screen.

Step 4—Determine level of confidence (HIGH or LOW) for each hazard level assigned

Level of confidence is determined by data source(s) and expert judgment of the overall strength of the evidence. Hazard classifications are represented in **BOLD** capital letters for high confidence. Hazard classifications are represented in *ITALIC* capital letters for low confidence. If a confidence level cannot be determined, then regular font is used.

Step 5—Assign a Data Gap to each hazard endpoint with insufficient information to assess

Assign a Data Gap (DG) classification to any hazard endpoints where there is insufficient information to assess the hazard using measured data on the parent chemical, measured data on a suitable analog, or estimated data on the parent chemical or suitable analog chemical.

Step 6a—Document findings and conclusions

It is essential to provide detailed documentation of the supporting data and rationale for all hazard classifications in an assessment report. The actual data values collected for each of the 22 chemicals are provided in the appendix Green Screen Data Collection.

Step 6b—Fill in the hazard summary table

Fill in the designated hazard classification level for each hazard endpoint in the respective box of the hazard summary table. The hazard summary tables are provided for each of the 22 chemicals in the appendix Green Screen Hazard Summary Tables.

Step 7—Determine the relative benchmark score

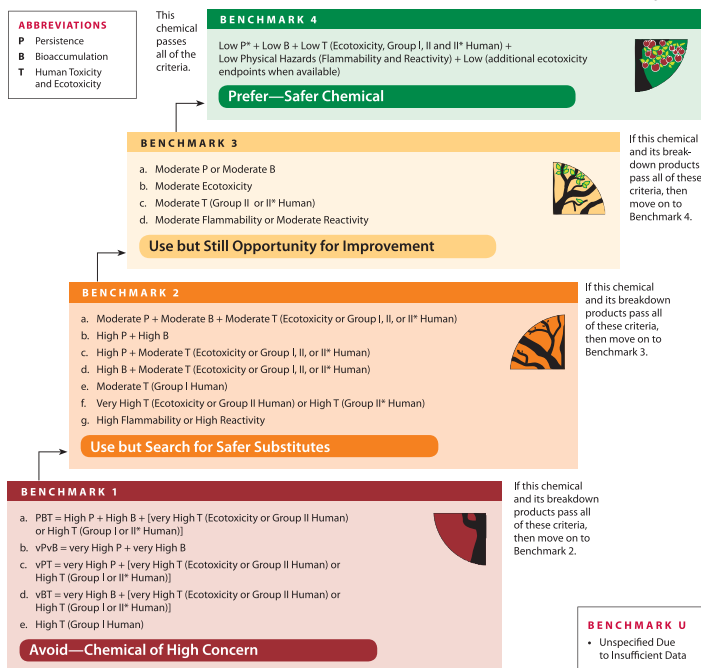
Using criteria established by Clean Production Action determine the relative benchmark score for each of the chemicals. The following are the four benchmark levels and are illustrated in [Fig. 1.3](#).

- Benchmark 1: Chemical of High Concern—Avoid.
- Benchmark 2: Use but Search for Safer Substitutes.
- Benchmark 3: Use but Still Opportunity for Improvement.
- Benchmark 4: Prefer—Safer Chemical.

OCTOBER 2011 (v2)

GreenScreen™ for Safer Chemicals v 1.2 Benchmarks

Start at Benchmark 1 (red) and progress to Benchmark 4 (green)



Group I Human includes Carcinogenicity, Mutagenicity/Genotoxicity, Reproductive Toxicity, Developmental Toxicity (incl. Developmental Neurotoxicity), and Endocrine Activity. **Group II Human** includes Acute Mammalian Toxicity, Systemic Toxicity/Organ Effects-Single Exposure, Neurotoxicity-Single Exposure, Eye Irritation and Skin Irritation. **Group II* Human** includes Systemic Toxicity/Organ Effects-Repeated Exposure, Neurotoxicity-Repeated Exposure, Respiratory Sensitization, and Skin Sensitization. Immune System Effects are included in Systemic Toxicity/Organ Effects. **Ecotoxicity** includes Acute Aquatic Toxicity and Chronic Aquatic Toxicity.

Note: The level of hazard indicated is the lowest hazard level at which a chemical would fail that criterion. However, if the chemical has a higher hazard level than what is listed (e.g. chemical is very High and the criterion is High), it would also fail that criterion.

* For inorganic chemicals with Low B, Low T (Ecotoxicity, Group I, II and II* Human) and Low Physical Hazards (Flammability and Reactivity), persistence alone will not be deemed problematic. Inorganic chemicals that are only persistent may achieve Benchmark 4.

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FIGURE 1.3 Criteria for Green Screen benchmark levels.*Step 8—Conduct a DG analysis to assign a final benchmark score*

Data requirements become more stringent with higher benchmark scores. With solid information on a single endpoint, one can confidently assess a chemical and assign a benchmark score of 1. Additional data are needed to assess a chemical and confidently assign it a higher benchmark score (e.g., Benchmark 2, 3, or 4). The number and type of DGs must be considered when assigning a benchmark score to a chemical. If a chemical does not achieve the minimum data requirements for the corresponding benchmark level, then it will be assigned a “U” for unspecified [16].

6.3 Environmental Protection Agency Design for the Environment

The EPA DfE [17] was developed to work with a wide range of stakeholders to help reduce the risk of chemical hazards to people and the environment. The reduction effort works toward preventing pollution by focusing on industries that combine the potential for chemical risk reduction and improvements in energy efficiency with a strong motivation to make lasting, positive changes.

The stakeholders include representatives from these industries as well as environmental groups. Working together, they have been able to develop goals and guidance for the work of the group. Partnership projects evaluate human health and environmental characteristics, performance, and cost of traditional and alternative technologies, materials, and processes. As incentives for participation and driving change, DfE offers unique technical tools, methodologies, expertise, and the potential for product recognition [17].

The DfE program focuses on promoting safer product development and using green chemistry alternatives through researched substitutions. The goal of this type of substitution is to reduce unintended consequences. To be considered safer choices, potential alternatives typically exhibit as many of the following characteristics as possible:

- They should be technically feasible
- Deliver the same or better value in cost
- Deliver the same or better value in performance
- Provide an improved profile for health and the environment
- Account for economic and social considerations
- Have the potential to result in lasting change [17].

The DfE Standard for Cleaning Products is intended to cover a wide range of products, including but not limited to: glass cleaners, general purpose cleaners, washroom cleaners, carpet cleaners, laundry detergents, graffiti removers, boat and car cleaning, drain cleaners, personal care, and floor care and other industrial products.

To be able to cover a wide range of product types, the DfE Master Criteria for Safer Ingredients (Master Criteria) was developed. It includes comprehensive, science-based criteria designed to ensure that the safest possible ingredients are used in the various DfE-labeled products. DfE looks at the ingredients of a product and not just the end product itself. This evaluation looks at an ingredient within its functional class context and on the chemicals key, distinguishing human health and environmental characteristics. By doing this, potential product ingredients can be viewed as part of a continuum of improved or safer ingredient choices.

The Master Criteria make it possible to set the spectrum for the greener or “low-concern” end of the continuum of chemical safety. To define low concern, DfE uses toxicological levels established by respected health and environmental protection authorities, including the United Nation’s GHS for the Classification and Labeling of Hazard Substances and the EPA’s New Chemicals

Program. For functional classes where no low-concern ingredients currently exist, DfE works with its stakeholders to carefully modify the Master Criteria in a way that allows for ingredient choices while ensuring the safest possible ingredients in that functional class. These criteria were designed for use in distinguishing safer chemicals for the DfE Program [18].

6.3.1 Acute Mammalian Toxicity

This category refers to those adverse effects occurring following oral or dermal administration of a single dose of a substance, or multiple doses given within 24 h, or an inhalation exposure of 4 h.

6.3.2 Carcinogenicity

Chemicals considered as carcinogens according to the authoritative DfE lists will not pass the criteria for DfE. Chemicals not on these authoritative lists, but that are known or presumed human carcinogens (Category 1), or suspected human carcinogens (Category 2) under GHS do not pass. Chemicals with limited or marginal evidence of carcinogenicity in animals do not pass the Master Criteria.

6.3.3 Genetic Toxicity

The more general terms genotoxic and genotoxicity apply to agents or processes which alter the structure, information content, or segregation of DNA, including those which cause DNA damage by interfering with normal replication processes, or which in a nonphysiological manner (temporarily) alter its replication. Genotoxicity test results are usually taken as indicators for mutagenic effects.

6.3.4 Neurotoxicity

This term means an adverse change in the structure or function of the central and/or peripheral nervous system following exposure to a chemical, physical, or biological agent.

6.3.5 Repeated Dose Toxicity

Repeated dose toxicity is evaluated using the GHS criteria for oral, dermal, inhalation of dust and vapors.

6.3.6 Reproductive and Developmental Toxicity

This reproductive category focuses on the occurrence of biologically adverse effects on the reproductive systems of females or males that may result from exposure to environmental agents. The toxicity may be expressed as alterations to the female or male reproductive organs, the related endocrine system, or pregnancy outcomes.

Developmental toxicity is concerned with the adverse effects in the developing organism that may result from exposure prior to conception (either parent), during prenatal development, or postnatal to the time of sexual maturation. Adverse developmental effects may be detected at any point in the life span of the organism. The major manifestations of developmental toxicity include: (1) death of the developing organism, (2) structural abnormality, (3) altered growth, and (4) functional deficiency.

6.3.7 *Respiratory Sensitization*

This category looks to reduce the use of a substance that will lead to hypersensitivity of the airways following inhalation of the substance.

6.3.8 *Skin Sensitization*

Likewise, this criterion limits the use of a substance that will induce an allergic response following skin contact.

6.3.9 *Environmental Toxicity and Fate*

Acute aquatic toxicity is the intrinsic property of a substance to be injurious to an organism in a short-term, aquatic exposure to that substance.

6.3.10 *Eutrophication*

Eutrophication is the process by which a body of water becomes enriched in dissolved nutrients (as phosphates), stimulating the growth of aquatic plant life usually resulting in the depletion of dissolved oxygen.

Table 1.5 lists more specifics on the various categories and thresholds for the DfE assessment.

In addition to the main ingredient requirements, the DfE program utilizes various life cycle considerations. Some of these elements include energy usage, nonozone-depleting substances and provide information on environmental, consumer, and worker safety matters. Further inclusions are that it restricts product volatile organic compound (VOC) content based on the most stringent government criteria and does not allow hazardous air pollutants or air toxics, except those that meet the DfE safer ingredient criteria. The program also does not allow products containing chemicals included on EPA's Toxics Release Inventory chemical list, except those that meet the DfE safer ingredient criteria. Ingredients must not exhibit the characteristic of ignitability and product pH must be ≥ 4 and ≤ 9.5 ; products with a pH outside this range may be considered for recognition if *in vivo* testing [18].

TABLE 1.5 DfE Master Criteria Cleaning Products^a [18]

Category	Thresholds	
Acute mammalian toxicity	Route (units)	Median lethal dose/concentration
	Oral, LD ₅₀ (mg/kg-bw) (by weight)	2000
	Dermal, LD ₅₀ (mg/kg-bw)	2000
	Inhalation, LC ₅₀ (vapor/gas) (mg/L)	20
	Inhalation, LC ₅₀ (dust/mist/fumes) (mg/L)	5
	R20: Harmful by inhalation R21: Harmful in contact with skin R22: Harmful if swallowed R23: Toxic by inhalation R24: Toxic in contact with skin R25: Toxic if swallowed R26: Very toxic by inhalation R27: Very toxic in contact with skin R28: Very toxic if swallowed <i>And all combination risk phrases containing one or more of the above (R20-R28)</i>	
	H300: Fatal if swallowed H301: Toxic if swallowed H302: Harmful if swallowed H310: Fatal in contact with skin H311: Toxic in contact with skin H312: Harmful in contact with skin H330: Fatal if inhaled H331: Toxic if inhaled H332: Harmful if inhaled	

Continued

TABLE 1.5 DfE Master Criteria Cleaning Products [18]—Cont'd

Category	Thresholds	
Carcinogenicity	National Toxicology Program	Known to be human carcinogen reasonably anticipated to be human carcinogen
	EPA	(2005/1999) Carcinogenic to humans, likely to be carcinogenic to humans, or suggestive evidence of carcinogenic potential -1 (1996) Known/likely (1986) Group A—Human carcinogen, Group B—probable human carcinogen, or Group C—possible human carcinogen-1
	IARC	Group 1—Carcinogenic to humans Group 2A—Probably carcinogenic to humans Group 2B—Possibly carcinogenic to humans -1
	EU CMR List	Category 1—Known to be carcinogenic to humans Category 2—Should be regarded as if carcinogenic to humans Category 3—Cause for concern for humans owing to possible carcinogenic effects
	EU Risk Phrases	R45: May cause cancer R49: May cause cancer by inhalation R40: Limited evidence of a carcinogenic effect And all combination risk phrases containing R45, R49, or R40
	EU CLP	H350: May cause cancer H350i: May cause cancer by inhalation H351: Suspected of causing cancer
	NIOSH Occupational Carcinogen List	http://www.cdc.gov/niosh/topics/cancer/npotocca.html
	GHS	Category 1A—Known to have carcinogenic potential for humans Category 1B—Presumed to have carcinogenic potential for humans Category 2—Suspected human carcinogens

Genetic Toxicity	EU CMR List	Category 1—Substances known to be mutagenic to man Category 2—Substances which should be regarded as if they are mutagenic to man Category 3—Substances which cause concern for man owing to possible mutagenic effects
	EU Risk Phrases	R46: May cause heritable genetic damage R68: Possible risk of irreversible effects <i>And all combination risk phrases containing R46 or R68.</i>
	CLP	H340: May cause genetic defects H341: Suspected of causing genetic defects
	GHS	Category 1A—Chemicals known to induce heritable mutations in germ cells of humans Category 1B—Chemicals which should be regarded as if they induce heritable mutations in the germ cells of humans Category 2—Chemicals which cause concern for humans owing to the possibility that they may induce heritable mutations in the germ cells of humans
Neurotoxicity	Route (units)	Median lethal dose/concentration
	Oral (mg/kg-bw/day)	100
	Dermal (mg/kg-bw/day)	200
	Inhalation (vapor/gas) (mg/L/6 h/day)	1
	Inhalation (dust/mist) (mg/L/6 h/day)	0.2
Repeated dose toxicity	Route (units)	Median lethal dose/concentration
	Oral (mg/kg-bw/day)	100
	Dermal (mg/kg-bw/day)	200

Continued

TABLE 1.5 DfE Master Criteria Cleaning Products [18]—Cont'd		
Category	Thresholds	
	Inhalation (vapor/gas) (mg/L/6 h/day)	1
	Inhalation (dust/mist) (mg/L/6 h/day)	0.2
	EU risk phrases	R48: Danger of serious damage to health by prolonged exposure (repeated exposure) <i>And all combination risk phrases containing R48</i>
	CLP	H372: Causes damage to organs H373: May cause damage to organs
Reproductive and developmental toxicity	Oral (mg/kg-bw/day)	250
	Dermal (mg/kg-bw/day)	500
	Inhalation (vapor/gas) (mg/L/6 h/day)	2.5
	Inhalation (dust/mist) (mg/L/6 h/day)	0.5
	EU CMR List-3	Category 1—Substances known to impair fertility in humans or known to cause developmental toxicity in humans Category 2—Substances which should be regarded as if they impair fertility in humans or cause developmental toxicity to humans Category 3—Cause concern for human fertility or possible developmental toxic effects

	EU Risk Phrases-3	R60: May impair fertility R61: May cause harm to the unborn child R62: Possible risk of impaired fertility R63: Possible risk of harm to the unborn child R64: May cause harm to breastfed babies R33: Danger of cumulative effects And all combination risk phrases containing R60-R64 or R33
	CLP-3	H360: May damage fertility or the unborn child H361: Suspected of damaging fertility or the unborn child H362: May cause harm to breastfed children
Respiratory sensitization	EU risk phrase	R42: May cause sensitization by inhalation
	CLP	H334: May cause allergy or asthma symptoms or breathing difficulties if inhaled
	GHS	Category 1A—High frequency of occurrence or sensitization rate in humans Category 1B—Low-to-moderate frequency of occurrence or sensitization rate in humans
Skin sensitization	EU risk phrase	R43: May cause sensitization by skin contact
	CLP	H317: May cause an allergic skin reaction
	GHS	Category 1A—High frequency of occurrence in humans and/or a high potency in animals Category 1B—Low to moderate frequency of occurrence in humans and/or a low to moderate potency in animals

Continued

TABLE 1.5 DfE Master Criteria Cleaning Products [18]—Cont'd

Category	Thresholds			
Environmental toxicity and fate	If a chemical is an acute aquatic toxicant (i.e., L/E/IC ₅₀ <100 ppm), then it must biodegrade rapidly and not be bioaccumulative (see lines 1-3 below). If a chemical has low aquatic toxicity (see line 4 below), then its half-life must be less than 60 days			
	Line	Acute aquatic toxicity value (L/E/IC ₅₀) 4,5,6	Persistence (measured in terms of level of biodegradation)	Bioaccumulation potential
	1	If ≤1 ppm...	...then may be acceptable if the chemical meets the 10-day window as measured in a ready biodegradation test without degradation products of concern8...	...and BCF/BAF <1000
	2	If >1 ppm and ≤10 ppm...	...then the chemical must meet the 10-day window as measured in a ready biodegradation test without degradation products of concern ...8	
	3	If >10 ppm and <100 ppm...	...then the chemical must reach the pass level within 28 days as measured in a ready biodegradation test without degradation products of concern8...	
	4	If ≥100 ppm...	...then the chemical need not reach the pass level within 28 days as measured in a ready biodegradation test if there are no degradation products of concern8 and its half-life <60 days...	
Eutrophication	The total level of phosphorus in the cleaning product will be limited to a maximum level of 0.5 wt% in the cleaning product as sold (measured as elemental phosphorus). Inorganic phosphates, as defined by the EPA New Chemicals Program, cannot make up any portion of the 0.5 wt% of phosphorus			

1. Chemicals listed as “possibly carcinogenic to humans” are evaluated largely on animal studies. DfE will consider appropriate data that show cancer concerns are not relevant to humans, e.g., because of an animal-specific tissue effect or mode of action. If the data demonstrate that cancer concerns are not relevant to humans, that chemical can be considered under the DfE Criteria
2. Per EU guidance, chemicals classified as Category 3 substances may be placed in that category based on positive results in assays showing (a) mutagenic effects or (b) other cellular interaction relevant to mutagenicity. If a chemical is classified in Category 3(b) only and that classification appears overly conservative, then the submitter may request EPA expert review. In such as case, if EPA determines the data do not support a concern for possible mutagenic effects, then the chemical will pass the criteria
3. The EU classification criteria do not currently consider a limit dose above which an adverse effect would not trigger classification. EPA will consider evidence demonstrating that a chemical carrying a reproductive/developmental toxicity risk phrase or listed as toxic to reproduction (in Table 1.6) did not cause an adverse effect below the Toxic Substances Control Act (TSCA) 8(e) Guidance Values listed in Table 1.6. Such a chemical may be determined, upon EPA review, to pass the DfE criteria for reproductive/developmental toxicity
4. In general, there is a predictable relationship between acute aquatic toxicity and chronic aquatic toxicity for organic chemicals, i.e., chemicals that have high acute aquatic toxicity may also have high chronic aquatic toxicity at low concentrations [20]. Since acute aquatic toxicity data are more readily available, the DfE Screens use these data to screen chemicals that may be toxic to aquatic life. Where measured chronic toxicity data is available, it will be assessed with other data and applied in the screen based on the relationship between acute and chronic aquatic toxicity
5. A case-by-case approach focusing on rate of biodegradation and degradation products of concern will be implemented for chemicals toxic to aquatic organisms at ≤ 1 ppm
6. For determining the aquatic toxicity of substances that are not toxic at their solubility limit, see ECOSAR Technical Reference Manual Figure 9, p. 17 (<http://www.epa.gov/oppt/newchems/tools/ecosartechfinal.pdf>); When a chemical may have effects at saturation as determined using the guidance in the ECOSAR manual, a weight-of-evidence approach in combination with US EPA expert review will be used. EPA may require additional testing including but not limited to solubility testing, chronic aquatic toxicity testing, or acute aquatic toxicity testing of analogs
7. Degradation products of concern are compounds with high acute aquatic toxicity ($L/E/IC_{50} \leq 10$ ppm) which mineralize $<60\%$ in 28 days

^aEU CMR, European Union carcinogenic, mutagenic and toxic to reproduction substances; CLP, Classification, Labeling, and Packaging; NIOSH, National Institute for Occupational Safety and Health (U.S.); ECOSAR, Ecological Structure Activity Relationships.

6.4 Green Seal™ Environmental Standard for Cleaning and Degreasing Agents (GS 34)

This environmental standard contains the basic requirements for cleaners/degreasers (as defined in its scope section) to be certified by Green Seal and for their manufacturers to receive authorization to use the Green Seal™ Certification Mark on products, on their packaging, and in product advertising and promotion. The requirements are based on an assessment of the environmental impacts of product manufacture and use and were developed with the guidance and input of a stakeholder committee consisting of solvent manufacturers, solvent users, government officials, academic and consulting experts, and environmentalists. The standard was reviewed and approved by consensus of the stakeholder committee, as defined by the International Organization for Standardization (ISO) and the Federal Government Office of Management and Budget Circular A-119. It is therefore a consensus standard. This standard is subject to revision as further experience and investigation may show it is necessary or desirable. Green Seal solicits information and advice on issues associated with this standard [19].

Requirements for the Green Seal™ standard are listed below.

6.4.1 Toxic Compounds

The product shall not be toxic to humans. An assumption is made that the toxicity of the individual component compounds is additive. A mixture need not be tested if existing toxicological information demonstrates that each of the ingredients complies.

6.4.2 Carcinogens and Reproductive Toxins

The product shall not contain any chemicals that are carcinogens or that are known to cause reproductive toxicity.

6.4.3 Corrosivity and Causticity

The pH of the degreasing agent concentrate shall be <11.0 but >2.5 , as measured using a pH meter and Method 9040 in Test Methods for evaluating solid waste, physical/chemical methods, EPA Publication, SW-846.

6.4.4 Skin and Eye Irritation

The degreasing product concentrate shall not be a skin or eye irritant.

6.4.5 Flammability and Ignitability

The concentrated product shall not be ignitable (i.e., the flashpoint for the compound is above 60°C (140°F)). In addition, the flash point of the final concentration of the degreasing product shall not be less than 5°C (40°F) above the manufacturer's recommended usage temperature.

6.4.6 *VOC Content*

VOCs include all organic compounds that have a vapor pressure of >0.1 mm Hg at 1 atm pressure and 20 °C. “VOC content” means the total weight of VOCs in a product expressed as a percentage of the product weight.

6.4.7 *Ozone Depletion*

The product shall not contain any ozone-depleting substances.

6.4.8 *Toxicity to Aquatic Life*

The product shall not be toxic to aquatic life. For purposes of demonstrating compliance with this requirement, the product shall be tested. However, aquatic toxicity testing is not required if sufficient aquatic toxicity data exist for each of the ingredients of the product to demonstrate that the product mixture complies.

6.4.9 *Aquatic Biodegradability*

The biodegradability of the finished product shall be determined using the protocols given in ISO Water-Quality Evaluation in an Aqueous Medium of the “Ultimate” Aerobic Biodegradability of Organic Compounds.

6.4.10 *Eutrophication*

Phosphates and phosphonates, including sodium salts and potassium salts, shall not be present in the product as used in quantities above 0.5% by weight of total phosphorus.

6.4.11 *Disposal*

The manufacturer shall either take back unused or spent products for recycling or disposal or provide the user with specific recycling and disposal instructions.

6.4.12 *Animal Testing*

To avoid new animal testing, previous test results will be accepted as evidence of meeting a criterion. When existing data are not available, the preferred methods for new testing include methods that replace, reduce, or refine animal use.

[Table 1.6](#) lists more specifics on the various categories and threshold values in the Green Seal™ standard.

6.5 **Quick Guide for Tools**

Cleaning product selection can benefit from a systematic comparison of health and safety topics. The four tools reviewed provide the end user with a couple of options. The two assessment tools can be used on a wide range of products but will require the end user to identify what these alternatives are. The two

TABLE 1.6 GS 34 Product-Specific Health and Environmental Requirements [19]

Category	Thresholds	
Toxic compounds	Oral LD ₅₀	<5000 mg/kg
	Inhalation LC ₅₀	<20,000 ppm of vapor or gas
		500 mg/L of mist, dust, or fumes
	Dermal LD ₅₀	<2000 mg/kg
Carcinogens and reproductive toxins	IARC	● Group 1 carcinogenic to humans ● Group 2A probably carcinogenic to humans ● Group 2B possibly carcinogenic to humans agents
	State of California under the Safe Drinking Water and Toxic Enforcement Act of 1986	Chemicals known to cause reproductive toxicity
Corrosivity and causticity	pH of the degreasing agent concentrate shall be	<11.0 but >2.5
Skin and eye irritation	Shall not be a skin or eye irritant	Considered a skin irritant if it has a mean score of 2 or more for either erythema and eschar formation or edema formation, based on the OECD dermal scoring system (OECD, TG 404)
		Eye irritant if it causes significant ocular lesions in any type of ocular tissue (i.e., cornea, iris, or conjunctivae) within 72 h after exposure that persist for at least 24 h
		Product concentrate shall meet all of the following criteria ● Mean score for cornea opacity of <2 ● Mean score for iris lesions of <1 ● Mean score for redness of the conjunctivae of <2.5 ● Mean score for edema of the conjunctivae (chemosis) of <2
Flammability and ignitability	Concentrated product shall not be ignitable	Flashpoint for the compound is >60 °C (140 F)

TABLE 1.6 GS 34 Product-Specific Health and Environmental Requirements [19]—Cont'd

Category	Thresholds	
Volatile organic compound (VOC) content	VOCs include all organic compounds with vapor pressure of greater than 0.1 mm Hg at 1 atm pressure and 20 °C	VOC content of the product as used shall not exceed the lower of the following options: ● 5% by weight ● The current regulatory limits of the Air Resources Board for the State of California (CARB) for its product category
Ozone depletion	Product shall not contain any ozone-depleting substances	
Toxicity to aquatic life	Compound is considered not toxic to aquatic life if it meets one of the following criteria	● Acute LC₅₀ daphnia or fish >100 mg/L ● Acute LC₅₀ algae >100 mg/L
Aquatic biodegradability	The compound shall meet the minimum requirement of	80% biodegradation based on dissolved organic content concentration
		70% of theoretical maximum CO ₂ production within 28 days
		Exception to this criterion can be made for products that are practically insoluble, i.e., solubility less than 10 mg/L
Eutrophication	Phosphates and phosphonates, including sodium salts and potassium salts, shall not be present in the product as used in quantities above 0.5% by weight of total phosphorus	
Disposal	Manufacturer shall either take back unused or spent products for recycling or disposal or provide the user with specific recycling and disposal instructions	
Animal testing	To avoid new animal testing, previous test results will be accepted as evidence of meeting a criterion	
	Preferred methods for new testing include methods that replace, reduce, or refine animal use, particularly those recommended by the Interagency Coordinating Committee on the Validation of Alternative Methods or the European Centre for the Validation of Alternative Methods, unless indicated otherwise	
	Other nonanimal (<i>in vitro</i>) test results, modeling data, data from structural analogs, and other lines of evidence may be accepted provided that the methods are peer-reviewed and applicable	

TABLE 1.7 Summary of Assessment Tools/Recognition Programs

P2OASys	Covers: worker, environment Ease of use: takes time to use; must find alternatives to compare; collect data Flexibility: applies to chemicals and equipment; can use with few end points Performance of product: not included
Green Screen	Covers: worker, environment Ease of use: takes time to use; must find alternatives to compare; collect data Flexibility: applies to chemicals Performance of product: not included
DfE Program	Covers: worker, environment Ease of use: chose from list of approved products Flexibility: chose from list, hard to compare to outside list Performance of product: included
GS 34	Covers: worker, environment Ease of use: chose from list of approved products Flexibility: chose from list, hard to compare to outside list Performance of product: included

recognition programs provide a list of products that have already been assessed for health and safety; however, if a comparison is desired with a product not appearing on the list, the end user must be able compare the products on their own. Only one of the four options allow for end users to consider equipment changes or improvements during their assessment process.

[Table 1.7](#) summarizes the intended use for each of the discussed tools/systems used in cleaning product selection, comparison, and evaluation.

Overall, each of the systems offers EHS criteria on many of the same areas of concern as summarized in [Table 1.8](#). Each system includes acute and chronic health effects, chemical/physical hazards, aquatic hazards, and environmental hazards.

7 PERFORMANCE TESTING/EFFICACY EVALUATION

If a product is green but does not work, then nobody is going to use it. The long running fear of green products has been tied to performance. Early on, this was a justifiable concern as many of the initial green products did not work as well under the same conditions as the traditional products. More often, extended cleaning times or additional forms of agitation were needed to ensure that proper cleaning levels were achievable. In an effort to address the shortcomings, independent performance testing was introduced to demonstrate that the next

TABLE 1.8 Summary of EHS Criteria

Hazards	P2OASys	Green Screen	DfE	GS 34
Acute human effects	Yes	Yes	Yes	Yes
Chronic human effects	Yes	Yes	Yes	Yes
Chemical/Physical hazards	Yes	Yes	Yes	Yes
Aquatic hazards	Yes	Yes	Yes	Yes
Persistence/ bioaccumulation	Yes	Yes	Yes	Yes
Atmospheric hazard	Yes	No	Yes	Yes
Disposal hazard	Yes	No	No	Yes
Energy and resource use	Yes	No	Yes	No
Product hazard	Yes	No	Yes	Yes
Exposure potential	Yes	No	No	No
Animal Testing	No	No	No	Yes

generation of green cleaners was as effective as conventional products and more effective than initial green products.

Solvents have been used for many years in all fields of cleaning. Solvent cleaning was often performed via single-species vapor degreasing operation. This streamlined approach was due to traditional solvents' superior ability to dissolve organic matter. In vapor phase cleaning, the final "rinsing" of the surface was accomplished by the condensation of solvent vapors.

As stated previously, many of these solvents, even though they work well, are regulated due to environmental and worker health issues. A new approach to the chemical selection process needs to be adopted to ensure that cleaning can be done in an effective, yet safe way. Identifying a substitute for solvents in cleaning applications is not an easy task. There are thousands of formulations to choose from. The literature for these products varies from vendor to vendor, as well as from product to product, making the search for an applicable substitute nearly impossible. Even after selecting a potential product, there is no guarantee that it will work.

The problem with the way most people evaluate cleaning product effectiveness is that their comparisons tend to be subjective, not scientific. However, well-written test protocols and repeatable science need to play a key role in choosing cleaning products that work consistently well. Part of the mission of the TURI Cleaning Lab is to test and evaluate the effectiveness of different cleaning chemistries and equipment on a variety of substrates and soils. The objective of the Lab is to help develop and promote environmentally friendlier,

safer alternatives to hazardous solvents. The TURI Lab has been aiding companies in the search for safer alternatives for more than 20 years and has performed thousands of cleaning trials.

Unlike these halogenated organic solvents, environmentally friendlier aqueous (i.e., water-based) detergents may not depend on their penetrability for their cleaning efficiencies, especially for the removal of petroleum-based surface debris. They rely instead on a number of chemical processes such as solubilization, wetting, emulsification, deflocculation, sequestration, and saponification [21]. Due to the variability in the alternative based products, performance is not guaranteed and therefore process-specific testing is needed.

7.1 Basic Testing Process

The basic outline of a cleaning test plan conducted at the TURI Lab follows a five step process (Fig. 1.4). Because of a decrease in the chemical energy relative to chlorinated organic solvents, several aspects of an aqueous cleaning system must be optimized prior to switching to the new cleaning formulation. These aspects include, but are not limited to, agitation methods, cleaner concentration, temperature, and cycle time.

7.1.1 Phase I Product Selection

Traditionally, this step included monitoring the effect of drops of selected detergent concentrates on grossly contaminated coupons over time. However, more often this step can be skipped due to the development of the CleanerSolutions database (www.cleansolutions.org). From the testing performed at the TURI Lab, data have been accumulated on the performances of industrial cleaning products. In order to use this information effectively, the CleanerSolutions database was created. This web-based interface is field-searchable by surface contaminants, surface substrates, cleaning equipment, solvents replaced, and vendor product data. The resulting list of alternatives helps companies to select products that are most likely to work for their specific situation.

7.1.2 Phase II Temperature and Concentration Studies

This phase and the next use the cleaning parameters of time, agitation, concentration, and temperature, collectively known as TACT. In the first round of scientific evaluation, surface cleaning analysis is conducted by measuring the weight of the artificially applied contaminant before and after cleaning. In addition, the source of agitation is minimized to focus the test on chemical activity. Cleaning time is kept constant to ensure accurate comparisons among the cleaners.

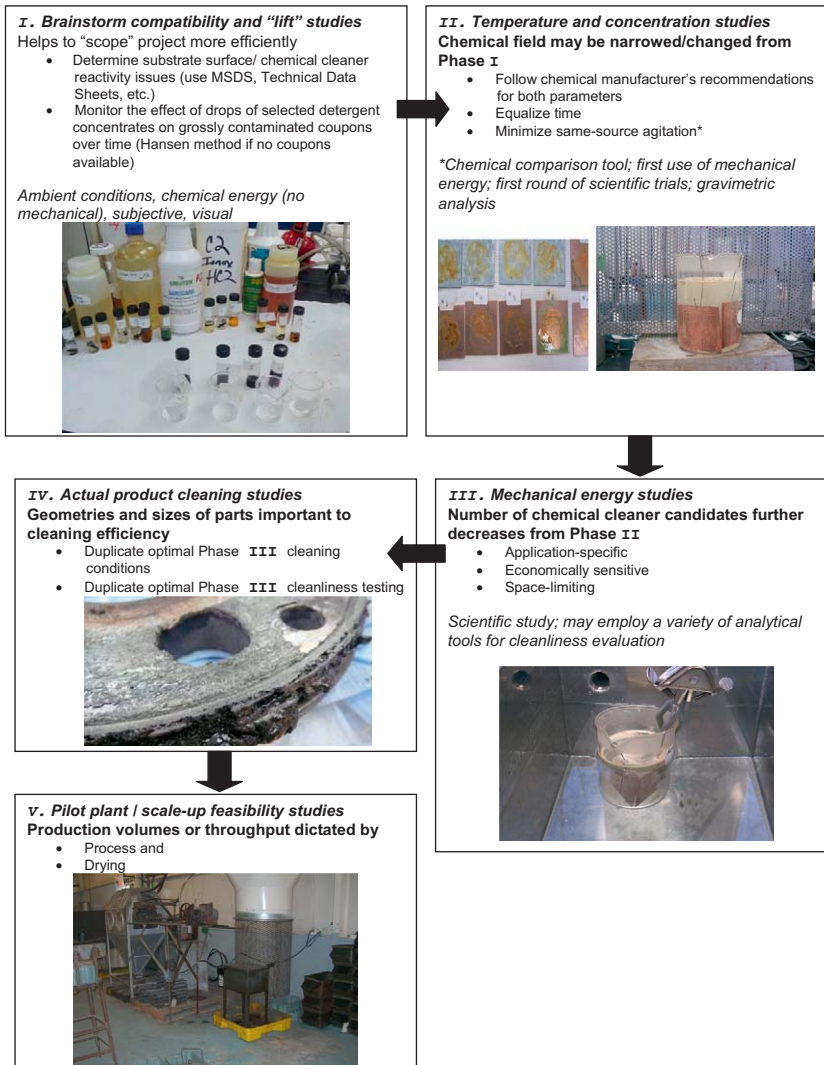


FIGURE 1.4 Cleaning phase of performance testing.

7.1.3 Phase III Mechanical Energy Studies

The third step is to enhance cleaning of the products from Phase II by introducing a form of mechanical cleaning. This can include manual scrubbing, ultrasonic energy, or spray washing. Artificially soiled coupons are again used to assess performance. Time, temperature, and concentration of the process can begin to be optimized during this phase.

7.1.4 Phase IV Actual Product Cleaning Studies

The next cleaning phase is concerned with the significance of part shape (screw configurations, blind holes, etc.) on metal cleaning. Note that in this model, mechanical energies (Phase III) are critically examined only after the proper cleaners, concentrations, and temperatures (Phase II) are ascertained. Other test protocols exist that may be as effective; it is essential that the experimental design be logical and consistent.

7.1.5 Phase V Pilot Plant/Scale-Up Feasibility Studies

During the on-site evaluation, a review of the top performing cleaning products is conducted to determine the potential impacts to the worker and/or the environment as well as the production process as a whole. During this on-site testing, it is important for the company to include workers involved with the cleaning process to participate in the piloting so they can ask questions and observe. In addition, input from the company's management is important to obtain to ensure final adoption will be financially acceptable.

With this basic procedure established, a wide range of cleaning solutions can be evaluated. The methodology prevents the need for a specific standardized test method for each individual scenario. The major industry sectors include metal working, electronics, chemical cleaning products, and general manufacturing (Fig. 1.5).

However, if a specific standard does exist, it can be incorporated into the TURI Lab process at Phase II or Phase III (Fig. 1.4), replacing the general procedure used during this phase of the evaluation process. Some existing standards that have been utilized for cleaning projects at the TURI Lab are listed in Table 1.9.

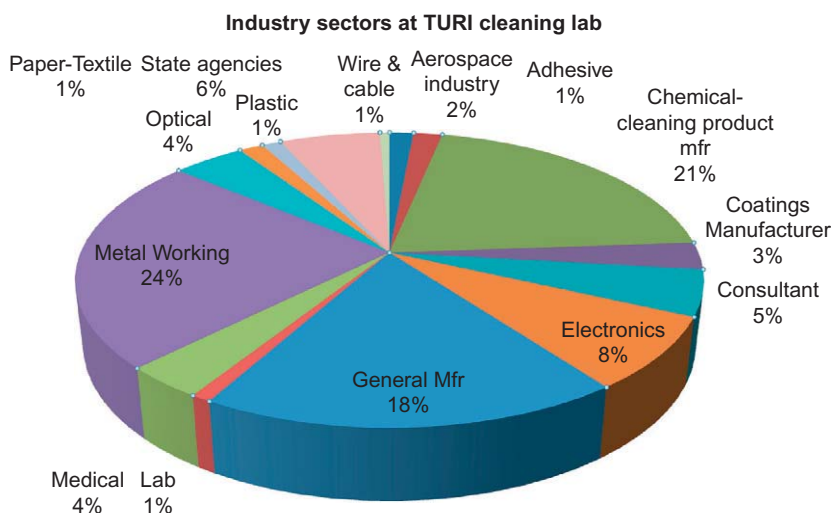


FIGURE 1.5 Industry sectors serviced at the TURI cleaning lab.

TABLE 1.9 Example Standardized Cleaning Protocols

ASTM	B322-85 Standard Practice for Cleaning Metals Prior to Electroplating A380 Cleaning, Descaling, and Passivation of Stainless Steel C756-87 Standard Test Method for Cleanability of Surface Finishes D 1133—Kauri-Butanol Value of Hydrocarbon Solvents D3938-96 Standard Guide for Determining or Confirming Care Instructions for Apparel and Other Textile Consumer Products D4010-83 Standard Specification for Waterless Hand Cleaner D4261-83 Standard Practice for Surface Cleaning Concrete Unit Masonry for Coating D6321-98 Standard Practice for the Evaluation of Machine Washable T-Shirts D6361-98 Standard Guide for Selecting Cleaning Agents and Processes D5703-95 Standard Practice for Preparatory Surface Cleaning for Clay Brick Masonry E1548-93 Standard Practice for Preparation of Aerospace Contamination Control Plans E1575-98 Standard Practice for Pressure Water Cleaning and Cutting E1857-97 Standard Guide for Selection of Cleaning Techniques for Masonry, Concrete, and Stucco Surfaces F502-93 e1 Standard Test Method for Effects of Cleaning and Chemical Maintenance Materials on Painted Aircraft Surfaces G31-72(1995)e1 Standard Practice for Laboratory Immersion Corrosion Testing of Metals G121 Preparation of Contaminated Test Coupons G122 Evaluating the Effectiveness of Cleaning Agents G127-95 Standard Guide for the Selection of Cleaning Agents for Oxygen Systems G131-96 Standard Practice for Cleaning of Materials and Components by Ultrasonic Techniques
Military/aerospace	MIL-PRF 85570 Performance Specification: Cleaning Compounds, Aircraft, Exterior MIL-PRF 87937 Performance Specification: Cleaning Compound, Aerospace Equipment MIL-PRF-680 Performance Specification: Degreasing Solvent MIL-PRF-29602A Performance Specification: Cleaning Compounds, Parts Washer And Spray Cabinet ARP 1755 Effect of Cleaning Agents on Aircraft Engine Materials Stock Loss Test Method
Third party certifiers and other standards	CCD-146 Hard surface Cleaners DFE standard for safer cleaning products Ecolabel CCD-110 Cleaning and De-greasing Compounds GS-41—Industrial and Institutional hand cleaners soaps GS-34—Green Seal Standard for Cleaning And Degreasing Agents GS-53—Standard Specialty Cleaning Products for Industrial and Institutional Use DCC-17—Greasy Soil Test Method for Evaluating Spray-and-Wipe Cleaners Used On Hard, Non-Glossy Surfaces DCC-15—Guidelines for Measuring Degree of Surface Abrasion by Abrasive Cleaners

7.2 Case Studies

To help demonstrate the complexity of switching to safer alternatives, the following three case studies from the TURI Lab have been included. Success comes by conducting systematic performance testing using the exact soils encountered at the facility and then to pilot the potential product on production parts by the personnel running the cleaning process. In addition, sometimes the solution to the cleaning problem may not be something traditional. There are projects where thinking outside the box can greatly improve the odds of success. The names of the cleaners and soils and contaminants, as well as the suppliers mentioned in the case studies, are listed in [Appendices A and B](#).

7.2.1 Case Study 1—*Shedding Some Light on Laboratory Versus Field Practices* [22]

This project focuses on how testing following standardized methods in the lab does not always translate into feasible alternatives in the real world. It emphasizes the need to work with those doing the cleaning in the field to ensure that success has been reached.

Summary—Assisted a manufacturer of solar panels in finding a safer way to clean parts during the assembly process. Worker health and safety was the initial driver for the company to want to find an alternative cleaning process. In addition, the company wanted to reduce their VOC emissions before opening a new, larger facility. The lab set out to find a replacement to manual wiping using isopropyl alcohol soaked towels during their final panel cleaning stage. The resulting product helped to improve drying the hands of workers and eliminated nearly 95% of their VOCs.

Information on Company—Solar Panel Manufacturer has been in business for more than a decade. The company made the decision when they opened to be a full service solar product manufacturer and to produce wafers, cells, and panels all in one facility. The company believes that being a fully integrated manufacturer allows them better quality control during manufacturing and assembly as well as promotes innovation and environmental responsibility in their process.

Current Solvent and Process—The existing process used towels that were manually soaked by the worker using isopropyl alcohol (IPA). In the past, the cleaning was performed with methanol. The IPA was substituted as the first attempt by the company to improve working conditions. The IPA-soaked towels were manually wiped across the surface of the solar panels until the worker felt no resistance or gripping during the wiping action. The intent was to remove any adhesive residue, fingerprints, or other foreign materials left over from the manufacturing process.

During the panel cleaning, the workers first visually inspect the back of the panel for any marks and wipe the backing of the panel with freshly IPA-wetted towels. Then they flip the panel over and look for any visible smudges on the panel and remove any gasket or adhesive seen on the panel or around the edges.

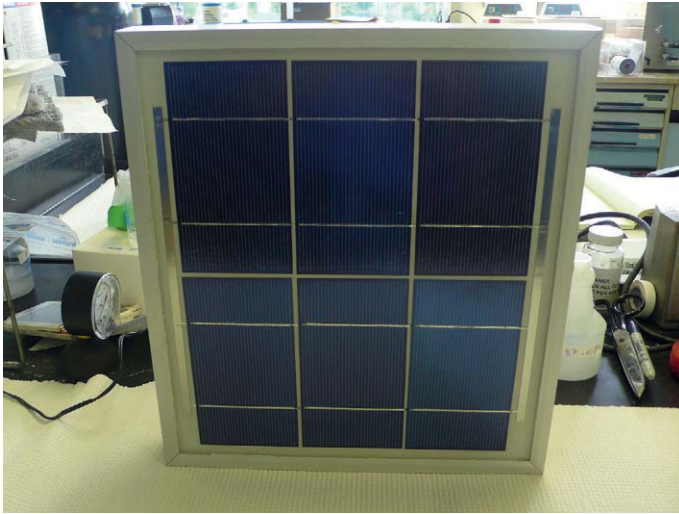


FIGURE 1.6 Solar panel assembly.

Materials to be Cleaned—The surfaces being cleaned were the final assembly of solar panels made up of aluminum housing, glass surfaces, and some laminated areas (back of panel). The main focus of cleaning was the glass surface. [Figure 1.6](#) shows a sample of an assembled solar panel.

Initial Contact—The lab was asked by the company to come and look at cleaning lines for a flux machine and at the end-of-line panel cleaning prior to packaging.

The lab performed a walkthrough of the facility to see the overall process and the two separate processes where the company is looking for substitute cleaners: flux removal and end-of-line panel cleaning. When products were found that perform satisfactorily for the cleaning process, the lab planned on going back on-site to test the products with the workers. This would allow for the collection of feedback on performance, odor, and reactions to skin.

Laboratory Testing—Adhesive, Ethylene Vinyl Acetate (EVA), and Fingerprint—Products were selected from the laboratory’s web database, www.cleanersolutions.org, based on supplied information from the client. Each product was used at full strength and room temperature. In addition, the current cleaning solvent, IPA, was included for comparative purposes.

Using a modified version of ASTM G122, the first round of testing was focused on the adhesive removal from glass. Prew weighed glass coupons were contaminated with a strip of the supplied adhesive tape. Coupons were weighed again to determine the amount of contamination added. Three coupons were then cleaned with each solution. Cleaning was performed by soaking a WypAll X60 reinforced paper towel with the cleaning solution. The coupons were then manually wiped for 2 min followed by a 10-s wipe with a dry towel. Visual

TABLE 1.10 Initial Lab Testing of Selected Alternatives

Cleaner	Average	Observations
Bio T Max	56.11	Removed top layer of adhesive; smudgy surface
Solsafe 245	74.04	Removed top layer and some of bottom layer; nonsmudged surface
Shopmaster RC	67.30	Removed top layer and some of bottom layer; nonsmudged surface
SC Actisolv	75.24	Good removal of top and bottom layer; nonsmudged surface
Optisolv OP 7171	5.09	No removal—top layer still sticky
Ionox HC 2	100.00	Excellent removal; required under 2 min (average 1:45)
DS 144	67.87	Good removal of top layer and some of bottom layer; nonsmudged surface
DS 800	−2.49	No removal—top layer still sticky; nonsmudged surface
IPA	63.77	Good removal of top layer and most of bottom layer; nonsmudged surface

observations were made and recorded during cleaning. Once dry, the coupons were weighed a final time and removal efficiencies were calculated for each product. Initial products tested are included in [Table 1.10](#).

The five products removing more than the current solvent were then evaluated under conditions more closely matching the current cleaning practice (residual adhesive removal only). In addition, odor comparisons were conducted to determine comparisons to IPA.

Laboratory Testing—Odor Evaluation—The eight selected products were compared with the current product, IPA, for odor levels. Each container of product was opened and wafted to determine the relative level of odor associated with the product. The products were ranked from worst to best by two members of the lab staff. The average rankings of the two were determined and reported. A normalized final ranking was made.

Many of the selected alternatives to IPA were found to have less objectionable odor based on the two-person panel. [Table 1.11](#) lists the general observational odor, the product rankings from lab staff members, the average odor rank, and the normalized ranking.

Laboratory Testing—Ethylene Vinyl Acetate Film Removal—During the follow-up trial, cleaning was performed on glass coupons that were contaminated with a strip of the supplied EVA supplied by the company. Cleaning

TABLE 1.11 Odor Observations and Rankings

Product	Observation	Rank 1	Rank 2	Ave. Rank	Normalized Rank
Bio T Max	Strong odor	9	9	9	9
Solsafe 245	Medium odor	7	7	7	7
Shopmaster RC	Strong odor	8	5	6.5	6
SC Actisolv	Low/no odor	4	2	3	2
Optisolv OP 7171	No odor	1	1	1	1
Ionox HC 2	Low/no odor	5	6	5.5	5
DS 144	Medium/low odor	3	3	3	2
DS 800	No odor	2	4	3	2
IPA	Medium odor	6	8	7	7

involved soaking a WypAll X60 reinforced paper towel with the cleaning solution and using it to manually wipe coupons for 1 min followed by a 5-s wipe with a dry towel. Visual observations were made and recorded during cleaning. Three of the alternatives and the IPA removed the EVA film in under a minute, with most requiring <30 s. One product removed the film from two of the three coupons. The IPA caused the EVA film to ball up on itself, whereas the other effective products caused the EVA film to be removed in one piece. Results are shown in [Table 1.12](#).

TABLE 1.12 EVA Cleaning

Cleaner	Observations
Solsafe 245	Slight removal
Shopmaster RC	Removed in under 30 s
SC Actisolv	Removed in under 20 s
Ionox HC 2	Removed in under 10 s
DS 144	Removed in under 60 s
IPA	Removed in under 60 s

The least objectionable odor products with the highest effectiveness were selected to clean the supplied part in the next stage of testing.

Laboratory Testing—Parts Cleaning—The top four products were selected from the previous trial. Each product was used at full strength and room temperature to clean one section of the supplied part. Cleaning was performed by soaking a WypAll X60 reinforced paper towel with the cleaning solution. The surface was manually wiped for up to 2 min followed by a 5-s wipe with a dry towel. Visual observations were made and recorded during cleaning. [Table 1.13](#) lists the observations made during cleaning of the front and back portion of the part.

Each of the four products was successful in cleaning residual dirt and fingerprints from the glass and aluminum surface. All had some difficulty removing the adhesive without first scraping the tape from the surface with a razor. The Ionox HC2 and SC Actisolv had the most success on the adhesive. In addition, based on visual inspection using the white towels, the lab decided that darker the wipes, the cleaner was the part. The same two products had the most dirt removal.

Laboratory Testing—Flux Removal Testing—Having identified several possible cleaning alternatives for the three main soils, the lab turned its efforts to the second project, removing flux from application equipment. Three products plus the current product were all used at full strength and room temperature for removing the supplied flux. Twelve preweighed aluminum coupons were coated with two layers of the rosin mildly activated flux using a handheld swab. Once dry, coupons were weighed a second time to determine the amount of flux added. Three coupons were manually wiped using the same method as the

TABLE 1.13 Visual Observations on Part Cleaning	
Cleaner	Observations
SC Actisolv	Good removal of dirt and fingerprints; adhesive softened but not removed. Backside cleaning removed second most amount as apparent by the coloring of the wipes after cleaning
Ionox HC 2	Good removal of dirt and fingerprints; adhesive softened more than the SC Actisolv but still not completely removed. Backside cleaning removed the most amount of dirt, fingerprints, adhesive as apparent by the coloring of the wipes after cleaning
DS 144	Acceptable removal of dirt and fingerprints; adhesive softened but still remained. Backside cleaning removed the least amount as apparent by the visual inspection of the wipes after cleaning
Shopmaster RC	Left residue behind—squeaks when wiped, some visual streaks. Backside cleaning removed third most amount as apparent by the coloring of the wipes after cleaning

previous trials, soaking the rag and wiping the surface. Coupons were wiped briefly with a dry wipe to remove any residual cleaner. Final weights were recorded and efficiencies were calculated for each coupon cleaned. [Table 1.14](#) lists the effectiveness of the selected products

All four products removed the flux from the aluminum coupons in under 5 min of manual wiping. For some of the cleaners, the efficiencies were determined to be over 100%. Typically, this can be due to residual soils that were on the surface prior to the initial weighing, or it can be due to solvents damaging the surface. In this case, there was no apparent visual damage to the aluminum coupons.

The company asked the Lab to evaluate the compatibility of top cleaning alternatives with the tin metal strips used in the manufacture of solar cells. This test was performed by using equal length strips of the supplied tin wire and placing them in a 10 mL glass vial containing a full strength product at room temperature. Each strip was weighed to establish a baseline weight. Strips were immersed in the solution for 24 h. The pieces were then removed and wiped to remove any residual cleaning product. Final weights were measured, and weight loss was determined. Observations were made during the 24-h period.

None of the products had any effect on the tin wire. There was no significant change in the weight of the strips over a 24-h period of immersion as shown in [Table 1.15](#).

TABLE 1.14 Flux Removal Results	
Cleaner	Average Cleaning Efficiency (%)
SC Actisolv	100.15
Ionox HC 2	97.67
DS 144	101.16
IPA	102.10

TABLE 1.15 Compatibility Test				
Product	Initial Weight	Final Weight	Change	% Change
SC Actisolv	0.0981	0.0980	0.0001	0.10
Ionox HC 2	0.1058	0.1059	−0.0001	−0.09
DS 144	0.0970	0.0966	0.0004	0.41
IPA	0.0904	0.0903	0.0001	0.11

On-site Piloting—With a few potential alternatives, the lab brought samples to the facility to provide on-site assistance.

The three products brought to the facility were tested in the TURI lab prior and were found to remove gaskets adhesive efficiently off glass coupons. The products were brought in 1000 mL bottles along with material safety data sheets. At the site, the solutions were used at 100% strength. Observations were made by the workers, EH&S and manufacturing staff as to performance, smell and drying rate of the products. Recorded observations from the piloting are listed in [Table 1.16](#).

TABLE 1.16 Pilot Testing Observations	
Product	Observations
Gemtek SC Actisolv	Did not slide across the panel like the IPA (grippy)
	Solution took the ink off the labels and bar code—not desirable
	Pretty good on the gasket removal—easier to remove than with IPA
	Better on hands than IPA. Observed some smell. IPA has a smell this was just different
	Drying time on the front of the panel took longer as it had less evaporation than the IPA
	Product will not be used in further testing
Dysol DS 144	Product moved well across surface
	It was alright on labels
	Pretty good at removing the gaskets; Product handled streaks well
	Odor was described as orange or citrus like. This was not likely to be an issue
	Product was slippery on hands
	Dried well (time) but it left residue
	Degradation test will be done in the lab to see if any harm is done to finished panel
	Product was left on-site for further piloting
Kyzen Corporation lonox HC2	Company informed the lab that this product was tried on the flux line, and while it cleaned the housing of the flux machine well, it was too tacky and left a residue so most likely it would not be something they would use in the shop

During the on-site testing, the company expressed concerns about VOCs. Unfortunately, the VOC levels of Dysol 144, the best product so far, would not lower the air emissions of company. In the new plant, the company will increase their use of cleaners by approximately eight times. Therefore, VOCs were a major concern. To address the new request, the lab would need to identify additional products and evaluate them following the initial laboratory testing process.

Laboratory Testing—Second Round of Testing—Seven products were selected from the lab’s Web database (www.cleanersolutions.org), based on information supplied from the client for target VOC levels. Each product was used at full strength and room temperature. Cleaning followed the same guidelines as the initial series of lab trials.

Laboratory Testing—Adhesive Cleaning Results—Three of the seven products removed more than 90% of the adhesive within 2 min of the manual wiping and three removed more than 60%. The Green Works Glass cleaner need <1 min to remove nearly all the adhesive. [Table 1.17](#) lists the cleaning efficiency for each coupon cleaned.

Laboratory Testing—EVA Film Cleaning Results—The top three products from the adhesive trial were used at full strength and room temperature. Only one product removed the EVA during manual cleaning. An additional product started to break the bond between the glass and the EVA but was not able to completely remove the contaminant from the surface. The Clorox Green Works Glass cleaner took <10 s to completely remove the film from the glass. Average effectiveness for the three products is listed in [Table 1.18](#).

Laboratory Testing—Flux Cleaning Results—Based on the success of the EVA film removal, only one product was evaluated for flux removal during the lab retesting process. The Green Works Glass cleaner worked well at removing the flux and could be used as potential cleaner for the flux operation. [Table 1.19](#) lists the efficiency of the Green Works Glass Cleaner.

TABLE 1.17 Retesting Adhesive Cleaning Results	
Cleaner	Average Cleaning Efficiency (%)
Bio Gold 3	91.32
Graffiti Remover SAC	63.29
Graffiti Remover	3.45
Soy Strong	90.60
Bean-a-doo	63.50
United 605	65.10
Green Works Glass Cleaner	99.28

TABLE 1.18 EVA Film Results	
Cleaner	Average Cleaning Efficiency (%)
BioGold #3	31.01
Soy Strong	0.95
Green Works Glass	99.99

TABLE 1.19 Flux Cleaning Results	
Cleaner	Average Cleaning Efficiency (%)
Green Works Glass	90.05

Laboratory Testing—Compatibility Study on Backing Material—The top three products from the previous follow-up trial and one product selected for on-site piloting were applied to a 7.62 cm × 7.62 cm (3 in. × 3 in.) square of supplied paper backing used in the manufacturing process. The cleaning products were allowed to remain on the surface overnight (18 h). Afterward, the squares were examined to determine if there was any visual damage to the material and if there was any cleaning product residue.

The previously selected product DS 144 and the Clorox Green Works Glass cleaner both had little to no residue remaining after sitting overnight. All four products did not cause damage to the paper backing. Observations are listed in [Table 1.20](#).

On-Site Piloting—Second Visit—The lab brought Clorox Green Works to the company’s facility for testing with the workers. At the site, the solutions

TABLE 1.20 Panel Backing Compatibility Observations	
Cleaner	Observations
DS 144	Only slight film; no visible sign of cleaner; no damage to surface
BioGold 3	Visual film on surface; slippery; no damage to surface
Soy Strong	Some film remaining but less than BioGold 3; some visual sign of cleaner; no damage to surface
Green Works	Only slight film, no visible sign of cleaner, no damage to surface

were used at 100% strength and transferred to squirt bottles, so the workers could use the solution in the same manner as IPA to clean the panels. Once the cleaning began, it was determined that a lesser amount may need to be used due to foaming of the cleaner on the panel. Using the spray delivery method instead of squirting may help the facility use less product per panel, keep foaming down, and give workers less exposure. Observations were made by the workers, EH&S, and manufacturing staff as to the usage, smell, and drying rate of the products. Observations are listed in [Table 1.21](#).

The Clorox product was retained for more on-site evaluations by the workers.

*Implementation—EH&S Comparison—*During the on-site evaluation, the lab reviewed the top performing cleaning product to determine the potential impacts to the worker and/or the environment. To do this, the lab used its comprehensive evaluation tool Pollution Prevention Options System (P2OASys). From the more comprehensive evaluation, similar results were found, with IPA having the most risk and the Clorox product having the least risk. [Table 1.22](#) lists the results from both EH&S evaluations.

TABLE 1.21 Second On-site Piloting	
Cleaner	Observations
Clorox Green Works Glass and Surface Cleaner	It did not slide across the panel like the IPA did. It was ok, but was not sliding as well as they were used to
	When the back of the panel was wiped with the product, the solution did not take the ink off the labels and bar code even with rubbing
	The worker said the product was pretty good on the gasket removal. The gasket was a little sticky but nothing they are not used to with the IPA
	The product felt slippery to the touch but it dried quickly enough to not be a problem
	Workers said the product did not smell at all and they felt they would have no problem using this product
	The product foamed a little when too much was put on the panel or a cloth was used repeatedly and product built up on the cloth. This will be a training issue that will need to be worked out on-site. Spray delivery may be better than soaking the cloth as was previously done
	Workers said the product worked and dried fine and that they could get used to the grippy nature of the product

TABLE 1.22 Environmental Health and Safety Comparison for Case Study 1						
Product		IPA	Green Works Glass	DS 144	SC Actisolv	Ionox HC2
P2OASys	Acute human effects	6	2	6	4	8
	Chronic human effects	6	2	2	2	4
	Physical hazards		2			
	Aquatic hazard	2				
	Persistence/bioaccumulation					
	Atmospheric hazard	2	2	2	2	2
	Disposal hazard	4	2	6	2	4
	Chemical hazard	9	5	8	9	8
	Energy/resource use	4	6	4	2	6
	Product hazard	6	2	6	2	2
	Exposure potential	6	2	6	6	6
P2OASys score (best = 2.0)		5.00	2.78	5.00	3.63	5.00

Follow Up—Next Steps—The desire of the company to find an alternative to IPA for cleaning was driven by two internal goals. The first goal was to improve worker health and safety by eliminating flammability concerns. The second goal was to reduce emissions of VOCs. The Clorox Green Works product satisfactorily met both of these goals as the product has no flash point and a VOC level of 40 g per liter is a nearly 95% reduction in VOCs. Perhaps most importantly, the product was found to work as well as the current cleaning process with IPA.

Since on-site piloting of the Clorox product, the lab has been notified by the company that there was hazing seen on the glass at their new facility and that customers had complained of hazing or etching on the glass. After researching the issue on their end, the company notified the lab that they would need to look for another cleaner for the glass that was more neutral. The Clorox product has a pH of 10.2 and the recommendations of the manufacturer are to use as neutral a cleaner as possible on the panel, no higher than 8.2. This fact was never disclosed to the lab at any time during testing, therefore this factor was not controlled during product selection.

The engineers at Solar Panel Manufacturer came up with the idea to look at cleaners for tinted auto glass since it is somewhat similar to the solar panels they produce. They are also more neutral in pH. A few additional cleaning products were found in the lab that met their pH requirements and were provided to the company to evaluate. All the new products were tested in the company's on-site environmental chamber to see if the etching or hazing still occurs with these lower pH products.

Working with the TURI Lab helped the Solar Panel Manufacturer to identify possible solutions to their panel cleaning. Even though the company opted to go with a different product that was identified during the testing, the interaction with the lab allowed them to recognize that there were many possible options to their process. The final identified options would maintain the current level of cleaning while lowering VOC levels and lowering worker exposure concerns.

7.2.2 Case Study 2—Thinking Outside the Cable Box—Wire & Cable Company Uses Floor Stripper on Cables [23]

Summary—A wire and cable company contacted the TURI Lab to help identify a safer cleaning practice for the removal of baked-on resin from steel wire guides used during the manufacturing process. The driving force behind the switch was a recent injury to a worker during the manual process which included chiseling and lathing. A Green Seal certified floor finish stripper was found to be the most effective cleaner when used at room temperature in both an ultrasonic unit and straight immersion bath.

Background—Information on the Company—Wire & Cable Company was established in 1926 and is a leading designer, manufacturer, and supplier of

armored cable, flexible wiring systems, and flexible conduit and fittings used in many types of buildings ranging from residential to office buildings. The mission of the company is to supply innovative, cost-effective, electrical, communication products, and systems that supply power, voice, and data for commercial, retail, institutional, and industrial applications.

Initial Contact—The lab was contacted for assistance in developing a new cleaning process because an employee was injured during a routine cleaning operation. The company paints and coats thin metal strips with paint.

Current Solvent and Process—The company previously used an acetone-based paint that evaporated readily leaving little residue on the guides that the metal cable or strips passed through.

When switching from the acetone-based paint to a water-based paint, the drying process was also changed. In the newer system, the cable is painted and run through a heating cabinet on guide. In this cabinet, the paint cures to a resin, consisting of dried paint and metal fines. The resin builds up on the guides or sheaths in the cabinet. As a consequence, the resins on the guides have to be cleaned periodically.

The time between cleaning the guides varies and the machine operator makes the decision to clean based on a visual inspection. The guides are approximately 15 cm in diameter, 1.27 cm thick with a groove in the guides where the cable runs. The guides are stainless steel and have a hole in the middle where the shaft goes through them. Some of the guides also have holes around the perimeter where the resin can gather. [Figure 1.7](#) shows a sample metal guide.

Before cleaning, the resin is cured. Where the resin adheres to the sides of the metal guide or is caught in the holes, the hard resin can be manually broken off. Generally, the resin is allowed to build up to about 3 mm but can be as much as 6 mm thick. This thickness would be considered to be a worst case scenario for this operation.

Cleaning of these parts was done by mechanical means; no chemicals were currently being used. The resin was lathed, hammered, and/or chiseled off. This



FIGURE 1.7 Metal guide.

is the process where an injury occurred. The company wanted to find another way to clean the parts that would decrease the risk of injury to their workers and still keep the EH&S profile of the cleaning operation as benign and nontoxic as possible.

The amount of time required for cleaning was not an issue as the company has replacement guides that can be put on a machine so the process can keep running. The company also was willing to buy equipment to clean the guides. The plan was to try to find the company a way to clean the parts with the following guidelines:

- Do not shift hazards on the worker (mechanical vs. chemical).
- Do not cause the company to have to pay a large amount for hazardous waste disposal.
- Do not use an excessive amount of energy.
- Do not use an excessive amount of manpower.
- Do not use an excessive amount of capital.

Laboratory Testing—Screening Test—An initial set of cleaning products was selected from the lab's web database, (www.cleanersolutions.org), based on information supplied by the client. Through this process, 21 potential products matched the search criteria for resin cleaning from steel parts. A 5 mL sample of each product at full strength was placed in a glass vial. A small piece of cured paint taken from a supplied part was placed in the vial and capped. The paint was allowed to sit in the solution to determine what effect the cleaning products would have. Observations were made at 10-, 20-, 30-, 45-, and 60-min intervals. Any signs of interaction between paint and cleaner were recorded. Observation summaries are listed in [Table 1.23](#).

Seven products that showed signs of dissolving the resin (color change and fizzing) were selected for further evaluation in the laboratory testing.

Laboratory Testing—Heated Immersion Testing—In addition to the top seven products, another seven products were identified using the labs online database, www.cleanersolutions.org, based on supplied information. During this evaluation, 100 mL of each product at full strength was placed in a 200 mL beaker. One product, Micro 90, was used at two dilutions, 10% and 20 %. A small piece of cured paint taken from a supplied metal guide was placed in the beaker and the beakers were placed in a basket and lowered into a large immersion tank that was heated to 49 °C.

The paint was allowed to sit in the solution to determine what effect the cleaning products would have. Observations were made at 15-, 30-, 60-, and 120-min intervals. Any signs of interaction between paint and cleaner were recorded. The solutions with the resin in them were then cooled and let to sit overnight. Observations were recorded the next morning. Summary of the products are listed in [Table 1.24](#).

All products showed some promise. Six cleaning products were selected based on performance, EH&S criteria, and potential costs to the customer.

TABLE 1.23 Screening Results for Resin Removal After 60 min

Product	Observations
Bio T Max	No change
Solsafe 245	No change
Citrus Burst 7	No change
Safe Strip	No change
DBE 6	No change
SC Aircraft & Metal Cleaner	No change
Micro 90 (10%)	Cloudy with some fizzing
Micro 90 (20%)	Cloudy with some fizzing
Ionox HC2	No change
Inproclean 3800	Solution darkened and had a lot of fizzing
VPW SC 1000	No change
Soysolv II	No change
Envirosolve	No change
Inproclean 4000 T	No change
D Greeze 500 LO	No change
Super Stripper	Solution darkened and had some fizzing
Enviro Star Green	Solution darkened and had some fizzing
Cycle Strip	Darkened, there was foaming and fizzing—clear dissolution of resin
EnviroCare Floor Stripper	Solution darkened and had some fizzing
Tar Remover	No change
Green Solutions	Solution darkened and had moderate fizzing

Laboratory Testing—Room Temperature Immersion Assessment—The six products from the previous trial were picked based on performance, EH&S criteria, and possible cost considerations. Hundred milliliter of each product at full strength was placed in a 200 mL beaker, except Micro 90 which was used at 10% and 20% dilutions. A small piece of cured paint/resin was placed in the beaker. The paint/resin was allowed to sit in the solution to determine what effect the cleaning products would have at room temperature. Again,

TABLE 1.24 Observations of Heated Immersion After 120 min

Product	Observations
Micro 90 (10%)	Some fizzing; some softening of the resin and solution had a slight tinting
Micro 90 (20%)	Fizzing and clouding; less softening than the 10% solution
Inproclean 3800	Softening, solution turning a darker color; a gel formed in the solution
Super Stripper	Slightly softened; solution separating into two parts, distinguished by color changes
Cycle Strip	Fizzing/foaming, some streaks of color; some softening
EnviroCare Floor Stripper	Fizzing; resin was very pliable; resin formed into a bubble shape
Green Solutions	Resin softened; solution amber colored; resulted in soft and rubbery feel
AK 225	Slightly softened
Vertrel XP 10	Softened; easy to pull resin apart
IPA	Slightly softened
HFE 7100	Slightly softened
OS 30	Slightly softened
Microcare MCA	Slightly softened
Ensolve	Initially softened resin; later formed bubble of resin 3 × size initial size; easily broken apart; no dissolution of resin as noted with no color change of solution; resin became brittle and could be broken apart

observations were made at 30-, 60-, 120-, and 180-min intervals as well as overnight. Any signs of interaction between paint/resin and cleaner were recorded. Observations are listed in [Table 1.25](#).

As was the case in the heated immersion trial, all of the products showed some promise. The Rochester Midland and Spartan products appeared to be top performers and among the top choices in the EH&S and economic categories. These two will be tested further in heated immersion and ultrasonic testing to find the optimum cleaning conditions.

Laboratory Testing—Parts Cleaning at Room Temperature—Cleaning took place using Rochester Midland and Spartan at full strength and room temperature. Circular steel parts approximately 15 cm in diameter with resin were immersed in the two solutions and were allowed to soak in the solution to

TABLE 1.25 Observations of Room Temperature Immersion

Product	Observations
Micro 90 (10%)	Slightly soft, brittle but still no dissolution and no solution color change
Rochester Midland EnviroCare Floor Stripper	Resin became very spongy; resin falls apart when touched; it seems to have expanded; no dissolution in solution
Spartan Green Solutions	Slight dissolution; solution brown in color but still clear; resin also expanded and spongy
Vertrel XP 10	Resin is brittle and could be broken apart; no dissolution
OS 30	Resin is brittle and could be broken apart; no dissolution
Microcare MCA	No dissolution; resin was softer

determine what effect the cleaning products would have on the resin. Observations were made at 60-, 120-, 180-, 240-, and 300-min intervals as well as overnight. Any signs of interaction between paint/resin and cleaner were recorded. A summary of the parts cleaning at room temperature results is listed in [Table 1.26](#).

Both the Rochester Midland and Spartan products worked. The cleaners soften the resin on the top which allows it to be wiped or scraped off in increments. These two were tested warm, at 49 °C, to determine if this would speed up the cleaning process.

Laboratory Testing—Parts Cleaning in Heated Immersion—The products from Rochester Midland and Spartan were again evaluated on the supplied parts by immersing the metal guides into the solutions. This time the immersion baths

TABLE 1.26 Observation of Room Temperature Cleaning of Supplied Parts

Product	Observations
Rochester Midland EnviroCare Floor Stripper	Product began to soften resin on part within 2 h and could be easily scraped off. The solution became darker over time. After overnight cleaning, resin could be wiped off
Spartan Green Solutions	Top layer of resin could be wiped off after 2 h. After 5 h, only thin layer of resin remaining on surface and could be wiped off easily

were heated to 49 °C. Observations were made at 60-, 120-, 180-, 240- and 300-minute intervals as well as overnight. Any signs of interaction between paint/resin and cleaner were recorded. Summary of the parts cleaning at the elevated temperature are listed in [Table 1.27](#).

Both the Rochester Midland and Spartan products softened the resin on the top of the part, which allowed the resin to be wiped or scraped off in increments. The difference between heating and room temperature cleaning was that the solutions blackened, the resin dissolved, and the resin seemed to soften faster with heat. Thinner areas of resin could be cleaned off in 3-5 hours rather than overnight. These products were tested with heated ultrasonic energy as described next.

Laboratory Testing—Parts Cleaning with Heated Ultrasonics—Test procedures were the same as the last trial except cleaning took place using a 40-kHz ultrasonic unit from Branson. The process concluded after 4 h of cleaning. Observations were made at 60-, 120-, 180-, and 240-minute intervals and are listed in [Table 1.28](#).

TABLE 1.27 Observation of Room Temperature Cleaning of Supplied Parts	
Product	Observations
Rochester Midland EnviroCare Floor Stripper	Solution darkened in color; resin seemed to be softening faster than when cleaned at room temperature; solution became very dark overnight
Spartan Green Solutions	Clear, fizzed, and saw streaks of color after 60 min; solution darkened quickly; resin softening, thinner areas down to metal; solution was black overnight; resin softening, thinner areas down to bare metal

TABLE 1.28 Observation of Room Temperature Cleaning of Supplied Parts	
Product	Observations
Rochester Midland EnviroCare Floor Stripper	Solution was dark brown almost black after first hour. Resin seemed to soften very quickly and peeled from the surface of the part. At the end of 4 h, the thicker areas of resin softened and could be peeled, scraped, and wiped off
Spartan Green Solutions	Solution was black and the resin softened within the first hour. After 4 h, the thinner areas were clear of any resin and the thicker resin softened and could be peeled, scraped, and wiped off

Both the Rochester Midland and Spartan products worked. The cleaners softened the resin on the top which allowed the resin to be wiped or scraped off in incremental layers. Using the heated ultrasonics reduced the cleaning time by approximately 50%. The resin dissolved in the solutions which may have a negative effect on the bath life. Thinner areas of resin were cleaned to the metal in approximately 2 h.

Recommendations would be to pull parts out and check the coating during the cleaning process. Parts should then be wiped every half hour with the added heat and energy in order to expedite the process of removing thicker layers of the resin/paint.

The next steps will be to supply the company with samples of the products and let the company test them to see which products they prefer and what arrangements they can make for routine cleaning. Since time was not a major issue in the cleaning process, the plan was for the company to try to work with room temperature immersion or ultrasonic cleaning; hence, the resin will not dissolve in the solution and possibly shorten bath life.

On-Site Piloting—Samples of the cleaning products that were evaluated successfully in the laboratory were delivered along with two five gallon ultrasonic tanks and two gallons of each cleaner for the facility to use and work with on their own for an extended period. They were told they had the option to heat or use the cleaners at room temperature.

They used the loaned equipment and samples for about a month and decided to go with the Spartan product at room temperature. They said they could soak the parts for hours or overnight and the removal was satisfactory.

The company was satisfied with the outcome of cleaning. Furthermore, they were pleased with the fact that the cleaner they selected was a Green Seal certified product which would allow for easy disposal and improve worker health and safety. After contact with the vendor of the product, the company was told that they could dilute and dispose of the solution down the drain. The resin was stripped off the parts in large pieces, which at room temperature allowed for easy separation of the resin from cleaning solution.

Implementation—The TURI lab returned to the facility to pick up the equipment after the piloting and to check on the status of implementation. The company bought a 35-gallon rectangular mobile polypropylene tank with a cover. They had a basket fabricated so that the workers could more easily lift the parts out of the tank. The tank is approximately 91 cm high by 91 cm wide and 30 cm deep mounted on casters. The tank was used for the soaking of parts. Their plan was to put a spigot about 15 cm from the bottom on one side, so the spent solution could be drained out and the resin could be collected and the two could be disposed of separately. They had already ordered a batch of the product and were ready to switch over the cleaning to the Spartan product.

The manager of the shop noted that the workers and management were happy with the new cleaning process overall. The Massachusetts location may eventually share the results with the rest of the company's facilities nationwide.

EH&S Comparison—During the on-site evaluation, the TURI Lab reviewed the top performing cleaning products to determine the potential impacts to the worker and/or the environment. To do this, the lab used P2OASys.

The current mechanical cleaning process was used to compare the two proposed alternatives using the P2OASys tool. Both chemical alternatives revealed that switching to either product would result in the potential for improved working conditions. [Table 1.29](#) lists the results from both EH&S evaluations.

Next Steps—The Company has implemented the change to the nontraditional product (floor stripping solvent) and is using the new chemical with success. They are using approximately 5 gallons a week in their room temperature immersion tank and are very happy with the system.

TABLE 1.29 Environmental, Health, and Safety Comparison for Case Study 2

Product		Mechanical Abrasion	Green Floor Finish Remover	EnviroCare Floor Stripper
P2OASys	Acute human effects	6	8	6
	Chronic human effects	2	6	2
	Physical hazards	10	4	4
	Aquatic hazard			2
	Persistence/bioaccumulation			6
	Atmospheric hazard	2	2	2
	Disposal hazard	6	2	4
	Chemical hazard	2	7	8
	Energy/resource use	6	4	6
	Product hazard	6	2	2
	Exposure potential	10	6	6
P2OASys score (best = 2.0)		5.56	4.56	4.36

7.2.3 Case Study 3—A Gear Company Decides to Go all the “Wayqueous”—Greasing the Wheels for Regulatory Change and Resistance from Above and Stepwise Replacement of Hazardous Cleaner [24]

Summary—A gear manufacturing company that had been working with the lab over the last decade had come to the realization that while the process may be difficult, the time to look at aqueous cleaning of its precision gearing parts had arrived. This was an important step in many ways for the company and the TURA program in MA. This decision showed that continued trust and work between state agencies and small MA companies could lead to the best overall TUR work and projects. This project has been one of several interim steps, working with issues of capital requirements, management support, and timing.

In the fall of 2007, the operations manager of this southern Massachusetts Gear Company called the TURI lab to ask about how to “get out” of TCE in their small vapor degreaser. The company had heard that the TURA program was going to list TCE as a high hazard substance and require anyone using 455 kg (1000 lbs) or more with 10 full-time employees to pay a higher fee and to report its use to the state, as well as develop a TUR plan. This company did not want to do this regulatory paperwork and reporting, in addition to its federal requirements for the EPA. Therefore, they wanted a drop-in replacement for TCE in their degreaser and bench-top applications as soon as possible.

Due to their immediate needs, the company was invited to the lab to bring parts for cleaning. The lab discussed with them aqueous and solvent cleaning options. The lab explained that there were many alternative cleaning chemicals that could be used for replacing TCE. The lab had worked with nearly two dozen companies from Massachusetts and other states in an effort to find less toxic alternatives to TCE. Often, companies were able to switch to an immersion cleaning process using an aqueous-based product. The lab’s field scientist typically found that cleaning was not the issue, but the rinsing and drying steps were the challenge.

The lab has conducted more than 100 experiments on 11 contaminant types from a wide range of industry sectors. A majority of the effective alternatives are alkaline aqueous products. Despite the success of finding aqueous-based alternatives for TCE, the products tested cannot be used in vapor degreasing due to their physical limitations. Likewise, these alternatives cannot be used where water would pose compatibility issues with certain processing applications.

The lab discussed with the gear company the solvent alternatives available at the time. The lab has investigated several solvents for replacing TCE specifically for vapor degreasing applications. The lab conducted multiple performance trials for six chemical classes on a variety of soils and substrates. The chemicals evaluated were hydrochlorofluorocarbon (HCFC), hydrofluorocarbon (HFC), hydrofluoroether (HFE), n-propyl bromide (nPB), volatile methyl

siloxanes (VMS), and dichloroethylene (DCM). Testing revealed that the nPB-based products were the only class to be effective on each of the 11 contaminants investigated as shown in [Table 1.30](#).

Based on concerns of the possible effects of water on parts, the company was very leery of the aqueous options. The company explained that its gears are very expensive and with the high surface tension of water and the small blind hole parts, aqueous cleaning might be too risky a change. The company asked about the drop-in solvents and more specifically about nPB.

Initial testing of the drop-in solvent was done at the lab with the gear company operations manager. The goal was to find a solvent that could be dropped into their small vapor degreaser. The main issues were that the cleaner needed to remove oils, metalworking fluids and metal fines from the surface of the part, as well as deep inside very small blind holes. This was important as the gears were not going to be plated. Therefore, if anything that could cause rust or corrosion was left on the surface or, if any fines were left in the blind holes as shown in [Fig. 1.8](#), the batch of parts could be rejected and contracts could be lost. Therefore, surface tension of the cleaner as well as solvency was a major issue. Cost also needed to be addressed to maintain current operating costs.

The lab had previously worked on a project in Rhode Island with the EPA Region 1 office. Drop-in replacement solvents for TCE and PCE (perchloroethylene) were evaluated and tested in the lab. The lab found that metal finishers wanted to move away from these toxic, traditional metal cleaning solvents due to regulations, but they also wanted to use their existing machinery to keep costs the same. Of the solvents tested, nPB was not only found to be the highest performing replacement solvent for TCE and PCE, but it was also the most affordable.

At the time, the EPA indicated that nPB was an acceptable replacement solvent. The chemical was SNAP (Significant New Alternatives Policy) approved due its lower environmental toxicity compared to TCE. Yet, at the same time, the American Neurological Society presented a small study in 2008 that showed the systemic neurological toxicity of nPB, specifically as a carrier in adhesives in furniture foam. Over the years that followed, nPB has been seen by many as a lateral move from TCE and PCE for use in cleaning. But nPB took hold as an alternative due to its lack of regulatory requirements, solvency, ease of transition, and cost.

What the Gear Company found during the switch to nPB from TCE was that they were using 25% less solvent. Thus, the company was producing 25% less hazardous waste. They were essentially doing TUR by reducing the amount of solvent used. In addition, costs were not much higher using nPB even though it was more expensive. The company was buying less solvent so the cost increase was not a major issue. Also, they were saving money on reporting their TCE use to the Commonwealth of Massachusetts and the federal government agencies, as well as paying less for waste disposal. The training to use the nPB was

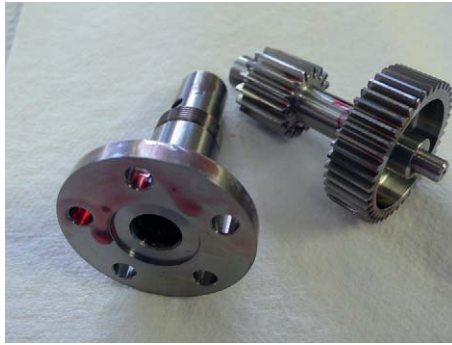


FIGURE 1.8 Gear system to be cleaned.

minimal. Unfortunately, EH&S was no better, but the company was willing to accept that trade-off.

When it became clear to the company that nPB regulations would become more costly, the company contacted the Lab to investigate aqueous solutions. The Lab has initiated research into aqueous options for the gear company and expects to have a solution in place within a 12-month period. The Lab's investigation will use the same testing processes and methodologies that the Lab has employed successfully for many other companies.

7.3 Case Study Summary

These three examples demonstrate how achieving success in reducing regulatory compliance by going to safer cleaning products and processes can be achieved. The first study illustrates how laboratory work is only part of the process and does not guarantee success. It reveals how the workers in the field must be included in any change in process.

In the second project, the wire and cable company worked with the Lab to uncover a nontraditional solution to the change in cleaning technology and, at the same time, reduced regulatory reporting requirements.

The third example demonstrates the value of incremental, iterative steps to a safer cleaning alternative.

8 SUMMARY

As stated at the beginning of this chapter, industrial surface cleaning is performed for a number of reasons. Because of the serious mechanical accidents that can occur as a result of poor surface cleaning and/or inspection, almost all cleaning regimens are developed on an application-specific basis. Historically, solvents have been used for many years in all fields of cleaning. Even though they work well, these chemicals pose health risks to workers and the

environment, therefore are regulated by state and federal agencies. As one tries to follow the vast list of regulations, identifying a substitute for solvents in cleaning applications is not an easy task. The alternative product selection process must be individualized to each scenario to ensure the best fit for the company conducting the evaluation. Success will only come by conducting systematic performance and then to pilot the potential product on production parts. Often, the solution to the cleaning problem may not be something traditional.

APPENDIX A CONTACT INFORMATION FOR CLEANING PRODUCTS AND SUPPLIERS

Product	Company	Location	Phone	Web Site
Bio T Max	Bio Chem Systems	Wichita, KS	800-777-7870	www.biochemsys.com
Solsafe 245	Bio Chem Systems	Wichita, KS	800-777-7870	www.biochemsys.com
Shopmaster RC	Buckeye International	Mayland Heights, MO	800-321-2583	www.buckeyeinternational.com
SC Actisolv	Gamtek Products	Pheonix, AZ	800-331-7022	www.gemtek.com
Optisolv OP 7171	Kyzen Corperation	Nashville, TN	615-831-0888	www.kyzen.com
Ionox HC 2	Kyzen Corperation	Nashville, TN	615-831-0888	www.kyzen.com
DS 144	Dysol	Fort Worth, TX	817-335-1826	www.dysol.com
DS 800	Dysol	Fort Worth, TX	817-335-1827	www.dysol.com
IPA	Fisher Scientific	Pittsburgh, PA	800-766-7000	www1.fishersci.com
Bio Gold 3	Vertec BioSolvents	Downers Grove, IL	630-690-0600	www.vertecbiosolvents.com

Continued

Product	Company	Location	Phone	Web Site
Graffiti Remover SAC	Spartan Chemical Company	Maumee, OH	800-537-8990	www.spartanchemical.com
Graffiti Remover	Finger Lakes Chemical	Rochester, NY	800-876-0222	www.castlepackpower.com
Soy Strong	Spartan Chemical Company	Maumee, OH	800-537-8990	www.spartanchemical.com
Bean-a-doo	Franmar Chemical	Bloomington, IL	800-538-5069	www.franmar.com
United 605	United Laboratories, Inc.	St. Charles, IL	800-323-2594	www.unitedlabsinc.com
Green Works Glass Cleaner	Clorox Co	Oakland, CA	800-292-2200	www.thecloroxcompany.com
Citrus Burst 7	Florida Chemical Company	Winter Haven, FL	863-294-8483	www.floridachemical.com
Safe Strip	Ecolink	Tucker, GA	800-886-8240	www.ecolink.com
DBE 6	Invista S.a.r.l	Wilmington, DE	800-231-0998	www.flexisolv.com
SC Aircraft & Metal Cleaner	Gamtek Products	Pheonix, AZ	800-331-7022	www.gemtek.com
Micro 90 (10%)	International Products Corporation	Burlington, NJ	609-386-8770	www.ipcol.com
Inproclean 3800	Oakite Products	Berkley Heights, NJ	800-526-4473	www.oakite.com
VPW SC 1000	Orison Marketing	Abilene, TX	325-692-1135	www.orisonmarketing.com

Continued

Product	Company	Location	Phone	Web Site
Soysolv II	Soysolv Industrial Products	Tiffin, OH	877-769-7658	www.soysolv.com
Envirosolve	21st Century Composites	Opa Locka, FL	772-834-3947	www.21stcenturycomposites.com
Inproclean 4000 T	Oakite Products	Berkley Heights, NJ	800-526-4473	www.oakite.com
D Greeze 500 LO	Solvent Kleene, Inc.	Peabody, MA	508-531-2279	www.solveentkleene.com
Super Stripper	Biokleen	Kalamazoo, MI	800-240-5536	www.biokleen.com
Enviro Star Green	Pioneer Eclipse Corp	Sparta, NC	800-367-3550	www.pioneer-eclipse.com
Cycle Strip	The Clean Environment Co	Omaha, NE	800-266-2353	www.cleanenvironmentco.com
EnviroCare Floor Stripper	Rochester Midland Corp	Rochester, NY	800-836-1627	www.rochestermidland.com
Tar Remover	United Laboratories, Inc.	St. Charles, IL	800-323-2594	www.unitedlabsinc.com
Spartan Green Solutions	Spartan Chemical Company	Maumee, OH	800-537-8990	www.spartanchemical.com
AK 225	AGA Chemical	Charlotte, NC	704-329-7614	www.ak-225.com
Vertrel XP 10	DuPont	Wilmington, DE	800-441-7515	www.dupont.com
HFE 7100	3 M	St. Paul, MN	877-666-2277	www.mmm.com

Continued

Product	Company	Location	Phone	Web Site
OS 30	Dow Chemical Company	Midland, MI	800-447-4369	www.dow.com
Microcare MCA	DuPont	Wilmington, DE	800-441-7515	www.dupont.com
Ensolv	Enviro Tech International	Melrose Park, IL	888-324-7400	www.ensolv.com

APPENDIX B CONTACT INFORMATION FOR SOILS AND CONTAMINANTS AND SUPPLIERS

Product	Company	Location	Phone	Web Site
Soltex Polyisobutylene	Soltex	Houston, TX	800-275-8580	http://www.soltexinc.com
Cargill, Inc. AP 60 Canola Oil	Cargill, Inc.	Minneapolis, MN	800-227-4455	http://www.cargill.com/
Nisseki SAS 40	JX Nippon Chemical Texas, Inc.	Pasadena, Tx	800-647-7354	http://www.jxncti.com
C.P. Hall Co Plashall ESO	HallStar	Chicago, IL	877-427-4255	http://www.hallstar.com
Castrol Industrial North America Castrol Quench G Quench Oil	Castrol Industrial	Naperville, IL	877-641-1600	http://www.castrol.com/
Rochester Midland Corp RI 780 Rust Preventive Coating	Rochester Midland Corp	Rochester, NY	800-836-1627	www.rochestermidland.com

Continued

Product	Company	Location	Phone	Web Site
Varnish Mix Cooper Creek Chemicals Cooper Black Tank Paint	Cooper Creek Chemicals	West Conshohocken, PA	610- 828- 0375	www. cooperscreekchemical. com
Exxon Mobil Vacmul O3D	Exxon Mobil	Houston, TX	281- 870- 6050	www.exxonmobil.com
Houghton International Cut-Max 135	Houghton International, Inc.	Valley Forge, PA	800- 666- 4684	http://www. houghtonintl.com/
Lanson Oil Co Vanish 6912	Lanson Oil Co.	Sarasota, FL	941- 351- 6704	
Houghton International Cindol 3401	Houghton International, Inc.	Valley Forge, PA	800- 666- 4684	http://www. houghtonintl.com/

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